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Master's Degree Program in
Data Science and Advanced Analytics

Fraud Detection Systems Empowered by Context-Awareness
Leveraging Dynamic Machine Learning Techniques

Daila Silva Alexandre

Master Thesis

presented as a partial requirement for obtaining a Master's Degree in Data Science and Advanced Analytics

NOVA Information Management School
Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

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by

Daila Silva Alexandre

Master Thesis presented as a partial requirement for obtaining the Master's degree in Data Science and Advanced Analytics, with a specialization in Data Science.

Supervised by

Supervisor: PhD Mauro Castelli, NOVA Information Management School

Co-Supervisor: PhD Nuno Rosa, NOVA Information Management School

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STATEMENT OF INTEGRITY

I hereby declare having conducted this academic work with integrity. I confirm that I have not used plagiarism or any form of undue use of information or falsification of results along the process leading to its elaboration. I further declare that I have fully acknowledged the Rules of Conduct and Code of Honor from the NOVA Information Management School.

Figueira da Foz, 09/07/2024

In honour of my mother and in memory of my father.

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ABSTRACT

In the dynamic and rapidly evolving field of financial technology, effective fraud detection systems (FDS) are crucial for maintaining security and customer trust. Traditional FDS methods often rely on static models, which struggle to adapt to emerging fraud patterns and evolving data distributions. This thesis explores the application of dynamic techniques to optimize fraud detection results, moving beyond conventional risk scores and static thresholds. The research employs a combination of machine learning (ML) approaches, including the use of secondary models for dynamic threshold adjustment and reinforcement learning (RL) for direct decision-making. A robust methodology was adopted, starting with an in-depth understanding of the business context, data exploration, feature engineering, and advanced sampling strategies to address data imbalance. Various strategies were tested, including anomaly detection using Isolation Forests, dynamic threshold setting through meta-learning, and RL to eliminate traditional score-based decision processes. The results demonstrated improvements in maximizing the true positive rate (TPR) while maintaining the false positive rate (FPR), ensuring more accurate and reliable fraud detection. By leveraging real-world data from a European transaction processor, this study provides empirical evidence of the benefits of dynamic optimization techniques in practical applications. The findings contribute to the ongoing development of adaptive and efficient FDS, highlighting the potential of combining advanced ML and RL methods to enhance the robustness and effectiveness of FDSs.

KEYWORDS

Fraud Detection; Fraud Detection Systems; Dynamic Systems; Machine Learning;
Reinforcement Learning

Sustainable Development Goals (SDG):



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LIST OF ABBREVIATIONS AND ACRONYMS

FD	Fraud Detection
FDS	Fraud Detection System
ML	Machine Learning
iForest	Isolation Forest
XGBoost	Extreme Gradient Boosting
LGBM	Light Gradient Boosting Machine
CatBoost	Categorical Boosting
RL	Reinforcement Learning

1. INTRODUCTION

The importance of effective fraud prevention and detection in the ever-evolving financial ecosystem cannot be overstated. Several fraud mechanisms have also emerged with the proliferation of diverse fintech solutions. Conventional fraud detection (FD) approaches are characterised by high costs, imprecision, and time consumption (Ashtiani & Raahemi, 2022). Therefore, innovative and intelligent methods aligning with modern financial transactions' intricacies are needed.

According to the European Central Bank Fraudulent Card Fraud Report (2023), transactions using cards issued in SEPA amounted to €1.53 billion in 2021, contributing to almost 0.03% of the total transaction value. This money obtained through fraud contributes to financing activities such as drug trafficking, organised crime, and even terrorism (Bhattacharyya et al., 2011). Additionally, a report by Javelin (2015) states that wrongful transaction declines due to fraud suspicion pose a significant threat to financial services, with an estimated €160 billion worth of declined transactions in 2014, affecting 15% of cardholders.

According to this information, a major challenge in the FD field is balancing the business goal of achieving a transaction acceptance rate compatible with a high-quality user experience while effectively detecting fraud (Alharbi et al., 2022; Mehbodniya et al., 2021).

In the real world, a fraud detection system (FDS) operates by assigning a risk score to each transaction, alongside an alert mechanism that triggers actions based on predefined thresholds. The alert system can be automatic or require manual evaluation by analysts. The fraud scoring component typically relies on a machine learning (ML) model, which is deployed in production.

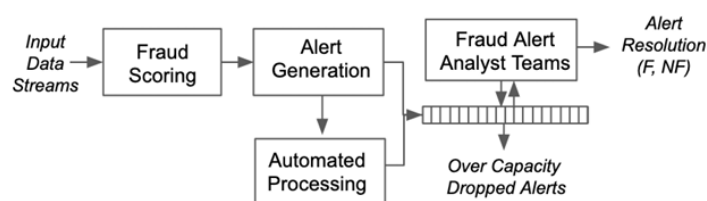


Figure 1 - Fraud Detection Process Flow (Shen & Kurshan, 2021)

One of the most challenging tasks is continuously monitoring these models already deployed in production, involving critical decisions about the potential decommissioning of the model and replacing it with an updated version.

Moreover, when optimising the model's threshold, the standard processes produce a static solution. However, in a dynamic environment such as fraud events, the ground truth for the fraudster's actions cannot be predetermined and integrated in advance. Hence, building an adaptive threshold to find the perfect balance between the true positive rate and the false positive rate using conventional methods is not feasible.

The threshold optimisation of a ML model for FD is intimately tied to some of the key business aspects:

- **Fraud Detection Performance:** The ability to avoid fraud losses constantly over time.
- **Operational Costs:** Depending on the alert action, the decision threshold will impact the contact center costs for client interactions and at the expense of the analyst team that verifies the suspicious transactions. This happens because the system goes through frequent periods of scoring decision inefficiency during which the analyst team may not have all relevant fraud cases in the alert population or true fraud alerts are disregarded due to a high volume of alerts/client contacts surpassing analysis/contact capacity.
- **User experience of the service:** Challenges in maintaining customer satisfaction arising from failing to prevent/block transactions for customers with true fraud cases or issuing alerts/blocking non-fraudulent transactions.

These aspects are challenging to quantify theoretically. However, a methodology can be proposed to balance mitigating fraud losses while preserving service quality significantly. The main objective is to find a dynamic threshold that maintains a false positive rate and a true positive rate over time.

Different segments may experience varying levels of fraud events over time. At times, the FDS's threshold might be set too low, leading to an excessive number of alerts for non-fraudulent transactions, degrading user experience, overloading the analyst team, and potentially causing relevant fraud cases to be missed. Conversely, if fraudsters frequently target certain segments, the system threshold might be too high, resulting in increased fraud losses and decreased customer satisfaction as actual fraud cases go undetected.

Furthermore, it is also possible to challenge the traditional approach of using risk scores and thresholds to decide whether to accept a transaction. This is where reinforcement learning (RL) comes in. RL, a type of ML where an agent learns to make decisions by performing actions and receiving rewards, can be applied to FD to adapt to new fraud patterns and improve decision-making processes dynamically. By its nature, RL is dynamic and allows for direct accept or reject decisions based on contextual data and the agent's learning. This method eliminates the reliance on predefined thresholds and leverages the RL agent's continuous learning capabilities to optimize real-time decision-making.

This research explores optimising a long-term FD model with recent data, avoiding the intensive process of retraining the entire model, which could risk losing relevant fraud memory. The approaches explored here aim to streamline the deployment process, monitoring, and canary strategy for assessing the readiness of the challenger model for final decision-making.

To make a dynamic system, a range of strategies were devised, such as employing a dynamic threshold and implementing RL techniques. These approaches included sophisticated sampling methods to address data imbalance, ensuring the system could effectively adapt to new fraud patterns. Each strategy was designed to refine the model's performance, maintaining robustness and accuracy in detecting fraud.

1.1. PROBLEM DEFINITION

The focus is on developing dynamical strategies to optimize FD, addressing the challenges posed by data drift, concept drift, and data imbalance, and ultimately improving the customer experience in financial transactions.

Research Questions:

- How can the impact of data drift and concept drift on the performance of DF models be mitigated?
- What sophisticated sampling methods can address data imbalance in FD, and what are their impacts on model performance?
- How can the threshold for FDS be dynamically optimized to balance the trade-off between false positives and false negatives effectively?
- How can a FDS be changed to eliminate the decision-making process based on risk scores and thresholds?

Research Objectives:

- To develop a robust approach for dynamically optimising the threshold of ML models in FDS using recent data.
- To implement and assess the effectiveness of RL techniques in enhancing decision-making processes in FDS.
- To design and evaluate sampling strategies to address data imbalance, incorporating the most reliable data and eliminating suspicious genuine transactions to improve model accuracy and adaptability.

1.2. STRUCTURE

This report is organised to provide a comprehensive overview of the research, from foundational theories to practical implementations and findings in the context of FD using ML techniques.

Section 2 - Literature Review: This section presents an extensive literature review, summarizing existing research and theories pertinent to FD. It examines various ML algorithms, sampling strategies, threshold optimisation techniques and RL techniques that have been used in previous studies. This review establishes the foundation for understanding the current state of the field and the gaps that this research seeks to fill.

Section 3 - Methodology: This chapter presents the theoretical background and framework of the study, detailing the key concepts, methodologies, ML and RL techniques employed. It outlines the process of model construction, including the selection and evaluation of algorithms, feature selection, and hyperparameter optimisation. Additionally, this section discusses the integration strategies when using ML techniques to dynamically optimize thresholds and the implementation of RL in the context of tabular data. The methodology adheres to best practices in ML and RL development, ensuring a robust and reproducible research process.

Section 4 - Results: This section presents the results of the study, including insights from data analysis and performance evaluation of the proposed strategies. The results are reported in terms of key metrics such as FPR, TPR, Precision, F1-Score, ROC curve, and AUC-ROC. The performance of different sampling strategies is also evaluated. A comprehensive discussion of the findings is provided, highlighting the effectiveness of the proposed strategies in optimising a FDS.

Section 5 - Conclusions: The final section synthesizes the research conclusions, reflecting on the overall contributions of the work and its implications for FD. It summarizes the efficacy of the strategies, the improvements achieved in classification performance, and the practical applicability of the proposed approaches. This section also addresses the challenges encountered, and the limitations of the study, and considers the potential for future developments and the broader impact of the research findings making recommendations for future research.

This structure seeks to provide a clear and logical progression from theoretical foundations to practical applications, ensuring a comprehensive understanding of research and its application in the FD field.

2. LITERATURE REVIEW

2.1. CONCEPT OF FRAUD

In simple terms, fraud refers to covert actions carried out by one party to gain an unfair advantage over another (Behdad et al., 2012). There are several types of fraud, including credit card fraud, insurance fraud, money laundering, and healthcare.

In the financial world, fraud can be defined as any form of illicit activity that exploits information from transactions with the intention of financial advantage (Awoyemi et al., 2017). More specifically, Irina Sakharova (2012) states that the use of personal data requires explicit consent for financial gain.

Modern fraud no longer needs fraudsters to be physically at the crime scenes. They can conduct their activities from a remote place, where they can conceal their identities (Alfaiz & Fati, 2022). This can be done through many devices such as credit and debit cards or online payment (Gates & Jacob, 2009).

According to Amiram et al. (2018), fraud in finance involves several components. The fundamental ones are (i) the presence of misrepresentation through misstatement, misreporting, or omission; (ii) the materiality of the misrepresentation; (iii) the person responsible for the act is aware of the crime; and (iv) in private legal actions, there is a causal relationship between the loss of the victim and the misrepresentation.

2.1.1. Difference between Fraud and Scam

It is important to recognize the difference between fraud and scam. Fraud encompasses a wide array of deceptive practices intended to gain an unfair advantage or cause harm, whereas scams focus on tricking individuals into revealing personal information or monetary assets (Button & Cross, 2017). It involves persuading individuals to participate in specific online activities for financial or malicious motives. One way to scam people is by engaging in fake online relationships that are designed to get the victims to transfer significant amounts of money over time (Williams et al., 2017).

This shows that in scams, the targeted individual plays an active role as the scammer capitalizes on their lack of awareness, whereas in fraud, the targeted individual assumes a passive role in the act since their involvement does not affect the outcome of the scheme. While scams serve as a more general term, fraud typically refers to actions that lean towards criminal or civil misconduct, often resulting in legal consequences (Button & Cross, 2017).

Despite this difference, the term "fraud" is used broadly to encompass both fraud and scams, as they will be treated as a single category throughout the research.

2.1.2. Causes and Consequences for Fraud

In this section, the causes and consequences of fraud will be explored, an aspect that is often overlooked in the existing literature. Understanding these factors may be important to comprehend the dynamics of the topic and for further analysis and discussion.

The Fraud Triangle is a conceptual framework proposed by Cressey (1953) that outlines three factors believed to be necessary for fraud to occur. The three elements of the Fraud Triangle are incentive, opportunity, and rationalization. The theory says that these components collectively create an environment conducive to fraudulent behaviour. Building upon this theory, the Fraud Diamond Theory, introduced by Wolfe and Hermanson (2004), adds a fourth factor, the element of capability.

The cardholder that suffers the attack is usually the entity that gets the least affected. This is due to current regulations, cardholder protection policies, and insurance programs in many countries that are designed to safeguard the rights of cardholders. However, the individual must invest time in bureaucratic processes. Conversely, merchants are significantly impacted since there is often no conclusive evidence to dispute a cardholder's claim of unauthorized card information use. As a result, merchants bear losses, including chargebacks, shipping expenses for goods, card issuer fees, and administrative costs (Quah & Sriganesh, 2008).

On a bigger scale, fraudulent action can lower public trust in the industry, impacting companies with deterioration of customer and stakeholder relationships; causing economic instability, and even impacting people's living costs (West & Bhattacharya, 2016).

2.1.3. Types of Fraud

In this section, three prominent types of fraud are explored. Unpacking these deceptive tactics empowers individuals and organizations to recognize and combat evolving cyber threats effectively.

2.1.3.1. Social Engineering Fraud

Humans are the weakest link in a security chain due to their vulnerability and natural inclination to trust (Krombholz et al., 2015; Salahdine & Kaabouch, 2019). Social engineering fraud is a type of manipulation that exploits this human psychological frailty to gain access to confidential information, systems, or funds (Atkins & Huang, 2013).

The types of social engineering attacks and their extent go as far as the fraudster's mind goes (Manske, 2000). In part, it can be considered a scam activity since it takes advantage of human weaknesses to make them carry out unsafe actions such as opening phishing emails (Williams et al., 2017).

2.1.3.2. Phishing

In the literature, there are many definitions for phishing, in any case, the important aspect to retain is that phishing is a cybercrime attack to obtain sensitive information by exploiting human vulnerabilities in electronic communication systems (Khonji et al., 2013).

Phishing mainly occurs when users are redirected to deceptive websites that usually look identical to legitimate websites (Varshney et al., 2024). This can be achieved through simple techniques such as sending a link by e-mail or SMS or through more complex techniques such as click-jacking which consists of misleading the victim into interacting with elements of the user interface from an alternate source, prompting actions unintended by the individual (Huang et al., 2012).

2.1.3.3. Malware

Malware, short for malicious software, is any software intentionally designed to harm or exploit computer systems or networks (Aslan & Samet, 2020). These programs make several attacks on security including compromising computer systems, pilfering sensitive information, disseminating spam emails, disrupting server operations, and more (Ye et al., 2017).

Malware programs can come in many forms, some of them being Viruses which are pieces of code capable of attaching themselves to other programs and infecting upon execution; Worms, unlike Viruses, can run independently, not needing another program to host; Trojan Horses that also run independently however, unlike Worms, they appear to be helpful programs but engage in harmful backend activity (Tahir, 2018).

2.1.4. Difference between Fraud Prevention and Fraud Detection

Fraud prevention and detection are distinct concepts with technical definitions. Fraud prevention is the first protection layer and involves halting fraudulent activities before they occur (Abdallah et al., 2016). To prevent fraud, one adopts practices that can be obtained through consumer education and transfer and payment processing companies can implement anti-fraud simple measures, such as real-time point-of-sale authorization, card verification codes, or more complex ones, for instance, the encryption of algorithms and firewalls (Abdallah et al., 2016; Bolton & Hand, 2001; Sakharova, 2012).

The second layer of protection, FD, is implemented when the first layer fails. FD focuses on recognizing fraudulent transactions as soon as they happen (Behdad et al., 2012). Commonly used mechanisms for FD include supervised ML algorithms for known target labels and unsupervised algorithms when these are not available (Ashtiani & Raahemi, 2022).

2.1.5. Fraud Detection Systems

A FDS is a comprehensive set of tools, processes, and technologies designed to identify, prevent, and mitigate instances of fraudulent activities within a given context, in this case, financial transactions.

A standard FDS comprises many layers of control, some of them automated and some under human supervision. Within the automated layers, it is employed various methods, including advanced algorithms, data analysis, and pattern recognition, to detect anomalies and unusual behaviour that may indicate fraudulent transactions (Carcillo et al., 2021). The human component plays a role when the system produces alerts for suspicious transactions. When the transaction is suspected of fraud, it is passed on to a specialized team for further investigation (Krivko, 2010).

As mentioned in the introduction, one of the most challenging tasks is the continuous monitoring of the algorithms that are in the automated layers. These algorithms are usually ML models and deciding whether to deactivate it and replace it with a new version is an important process in the monitoring. This process includes the following steps:

- Analysing both data drift and concept drift, indicating that the trained data may not properly represent fraud dynamics (Dal Pozzolo et al., 2015);
- Based on the outcomes of the drift analysis, deciding whether to train a new model with the same features or develop a new one with new features.
- Creating the deployment strategy, including validation out-of-sampling testing or shadow testing and canary deployment.
 - **Validation and out-of-sample testing**, particularly if a significant time gap exists between the training period and when the model becomes operational. This gap could introduce data drift, potentially impacting the optimisation of the classification threshold to reflect current reality.
 - **Shadow testing**, if fraud labels are accessible or if proxies can be constructed to classify potential fraudulent events (such as historical transactions from cardholders who have cancelled their cards), the threshold can be optimized using real-time data production or an A/B testing on specific segments of the population.

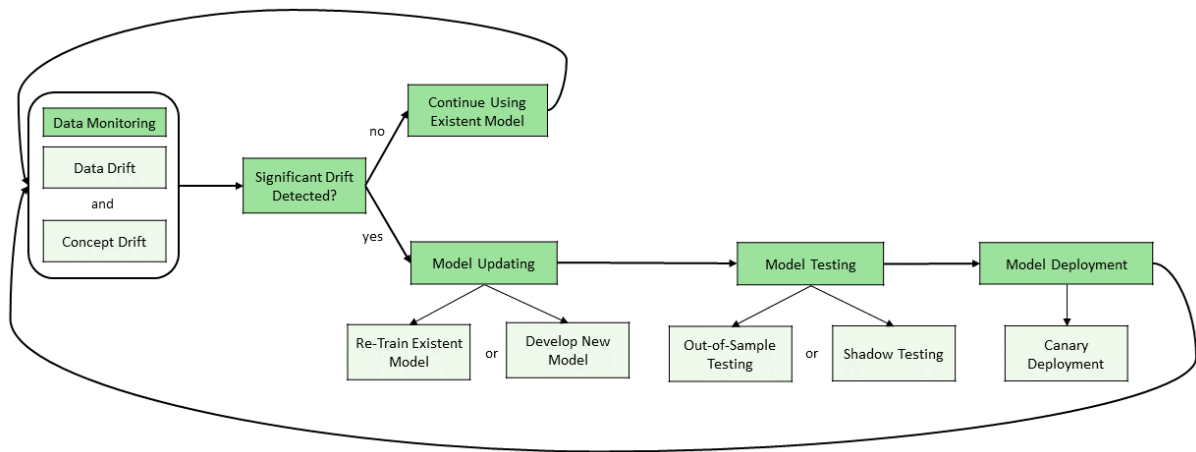


Figure 2 - Training Cycle of a Machine Learning Model in Production

In essence, a FDS based on a ML model demands ongoing updates to use the most recent labels or discard outdated information that might mislead and increase the False Positive Rate.

Besides the complexity of the operating systems, addressing the challenge of checking all transactions is also necessary. Every FDS relies on human labour to analyse and contact clients identified by the model. Investigators execute a validation process by reaching out to cardholders and then providing feedback to the FDS, indicating whether the transaction target by the model was fraudulent or genuine.

However, the costs associated with this process limit the number of alerts that investigators can validate. Given the current human capacity, the feedback can only refer to a tiny fraction of the daily transactions and serves the purpose of training and updating classifiers almost in real time. The labels associated with the remaining transactions experience a delay, as they depend on a customer reaction time. This means that all transactions not reported as fraud within this time frame are considered genuine, introducing noise to the class labels, and not reflecting the ground truth. The impact of this issue on model performance worsens in imbalanced datasets (Folleco et al., 2008).

2.2. ANOMALY DETECTION

Anomaly Detection, also known as anomalous event identification, aims to recognize uncommon or unexpected occurrences in data streams (Nassif et al., 2021; Schneider & Xhafa, 2022). These unexpected occurrences, also known as anomalies, are sometimes described as outliers - objects that stand out from others - but, ultimately, they are events that do not conform to the norm (Kotu & Deshpande, 2019). It is essential to first understand the norm to identify the events that are not the norm (Schiller et al., 2007).

Anomaly detection is crucial in various applications, especially in real-time scenarios, where the timely identification of anomalies is fundamental as in the case of FD (Schneider & Xhafa, 2022).

The classification of anomaly detection highlights three primary classes: unsupervised, semi-supervised, and supervised, with temporal trends delineating their popularity over different periods (Nassif et al., 2021). The choice depends on the data and the presence of labels. Supervised anomaly detection relies on a labelled training dataset to build predictive models that distinguish between normal and anomalous instances.

On the other hand, Unsupervised anomaly detection does not require labelled training data, making it highly applicable as it assumes normal instances are far more frequent than anomalies, and identifies outliers directly from the data. Semi-supervised anomaly detection builds a model using only labelled normal instances, identifying anomalies as instances that do not fit this model. This approach is more applicable than supervised techniques since it does not require labelled anomalies, and can be adapted to operate in an unsupervised mode by using a sample of the unlabelled dataset for training (Chandola et al., 2009; Yepmo et al., 2022).

2.2.1. Isolation Forest

Liu et al. (2008) proposed the Isolation Forest (iForest) as an unsupervised model to explicitly single out anomalies by creating an ensemble of Isolation Trees (iTree). The idea is based on two main properties of anomalies: they are a minority, meaning there are fewer of them, and their attributes are very different from those of normal instances. With this, anomalies are recognized as instances that tend to be isolated closer to the root of the tree.

iForest operates on the principle that anomalies can be effectively separated by recursively partitioning the data using a tree structure. This process results in shorter paths for anomalies for two primary reasons: first, shorter paths mean fewer partitions, which isolate anomalies faster due to their infrequency, and second, instances with significantly different values tend to be split early in the partitioning process.

Below, it is possible to visualize the operation. Anomalies are represented by shorter paths (depicted by a red line), while regular points take longer paths to reach the tree's maximum depth (depicted by a blue line), as shown in the following image.

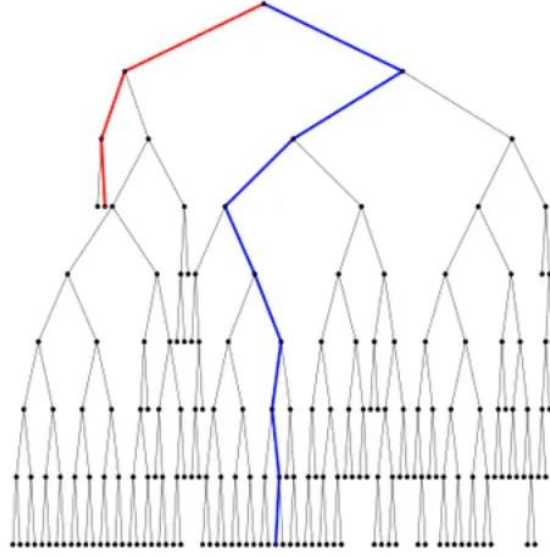


Figure 3 - Single Tree in a Forest (Hariri et al. 2021)

The anomaly score for an instance is calculated using the following formula:

$$s(x, n) = 2 \frac{E(h(x))}{c(n)} \quad (1)$$

Where $E(h(x))$ is the average path length of an instance x across all trees, and $c(n)$ is a normalizing factor representing the average path length of an unsuccessful search in a Binary Search Tree. Consequently, higher scores indicate higher anomalies. For this use case, instances are defined as anomalous based on a predetermined threshold (e.g., the highest 5% of scores).

This algorithm has a linear time complexity and a low memory requirement, making it highly efficient for large datasets. This is achieved through sub-sampling and building partial models, which significantly reduces computational costs compared to other methods. However, it introduces randomness, leading to lower accuracy and stability issues (Ding & Fei, 2013; Xu et al., 2017).

2.2.2. Extended Isolation Forest

Hariri et al. (2021) introduced the Extended Isolation Forest (EIF) as an extension of the existing iForest. This extension addresses specific shortcomings in the original algorithm, particularly related to the assignment of anomaly scores and the handling of high-dimensional data.

The primary motivation behind EIF is to improve the reliability and accuracy of anomaly detection. The standard iForest algorithm has a known bias due to its method of branching, which relies on axis-parallel cuts. This bias can result in artifacts in the anomaly score distribution, leading to inconsistent and unreliable scores. For instance, anomalies might be

misclassified due to the rectangular nature of the branching, which does not adapt well to the actual data distribution. On the other hand, EIF introduces the use of hyperplanes with random slopes for splitting the data. This approach helps in reducing the bias introduced by the standard method and allows for a more accurate representation of data anomalies.

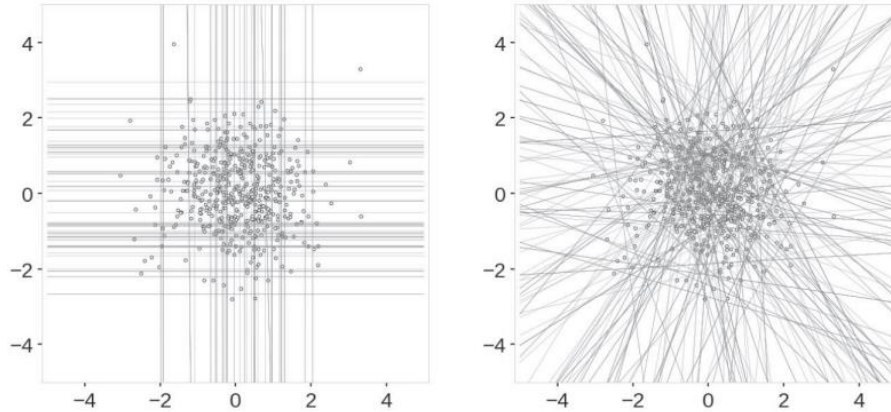


Figure 4 - Comparison of Isolation Cuts in iForest (left) vs. EIF (right) (Hariri et al., 2021)

EIF is designed to handle high-dimensional datasets effectively. The use of random hyperplanes allows the algorithm to perform well even with a large number of irrelevant features, maintaining computational efficiency with a linear time complexity and low memory requirements. The anomaly score in EIF is calculated in the same manner as the standard iForest.

Another enhancement involves transforming the data randomly before constructing each tree. This technique helps average out the bias across multiple trees, further improving the robustness of the anomaly scores.

The EIF significantly improves upon the standard Isolation Forest by addressing key limitations and enhancing accuracy and robustness. These advancements make EIF a powerful tool for anomaly detection, especially in complex and high-dimensional data environments. Its ability to produce reliable anomaly scores without sacrificing computational efficiency makes it suitable for various real-world applications.

The EIF significantly improves upon the standard iForest by addressing key limitations and enhancing accuracy and robustness. These enhancements make EIF a powerful tool for anomaly detection in complex and high-dimensional data environments, producing reliable anomaly scores without sacrificing computational efficiency, and making it suitable for a wide range of real-world applications.

2.3. TECHNIQUES FOR FRAUD DETECTION

Here are covered techniques for several types of financial FD. However, the literature places greater emphasis on credit card FD compared to other categories. (Bolton & Hand, 2001; Carcillo et al., 2021; Rodrigues et al., 2022).

It is worth mentioning that achieving a minimal level of fraud comes with a trade-off between heightened security measures and the associated costs and effort. While it is theoretically possible to block all transactions and eliminate fraud entirely, such an approach would impede regular business operations, resulting in a significant loss of customer satisfaction. Striking the right balance involves implementing effective fraud prevention measures without compromising the overall user experience and operational efficiency (Bolton & Hand, 2002).

In the past, simpler techniques were sufficient to capture deceptive transactions, relying on checklists that flagged potential signs of fraud. For instance, these methods would search for patterns such as frequent card declines, numerous unsuccessful attempts to input a credit card number, or instances where an individual exceeded their financial means (Chaquet-Ulldemolins et al., 2022b).

However, since fraudsters are always improving their techniques, FDS must be flexible enough to change with them (Al-Hashedi & Magalingam, 2021; Bolton & Hand, 2001). Either way, new criminals entering the field may not know about past successful detection methods, leading them to use tactics that can be identified as fraud by old-fashioned approaches. This highlights the need to apply earlier and latest detection tools (Bolton & Hand, 2002).

The latest techniques involve ML, as these methods produce results with higher accuracy and precision, have better computational efficiency, reduce costs, and improve customer satisfaction (Chaquet-Ulldemolins et al., 2022b). Additionally, Data Mining approaches prove efficient due to their capacity to detect subtle anomalies within extensive datasets (West & Bhattacharya, 2016). Besides ML and Data Mining, FD employs approaches from a variety of other disciplines such as mathematics, statistics, engineering, and artificial intelligence (Al-Hashedi & Magalingam, 2021; Ruff et al., 2021).

Within the context of ML, the literature is essentially divided into supervised learning and unsupervised learning. Each approach has its benefits and limitations which will be discussed in the respective sections. Recently some authors have suggested combining the two techniques stating that they complement each other and lead to better results (Carcillo et al., 2021; Micenková et al., 2014).

2.3.1. Heuristic Rules vs Machine Learning

The debate between heuristic rules and ML algorithms is a critical consideration in the realm of FD. While the sophistication of ML models provides a robust framework for identifying complex patterns indicative of fraudulent behaviour, there are instances where static rules, often referred to as heuristic rules, can play an essential role.

These heuristic rules are built in the form of If {assertions}, Then {consequence} and are, usually, derived from expert knowledge or historical patterns. They offer a quick and deterministic approach to flagging potential fraud and are quite often used to that effect in the banking sector (Krivko, 2010).

2.3.2. Data Split

Detecting fraud presents a multifaceted challenge due to its complex nature, as fraudsters continuously devise new tactics to commit fraud and avoid detection, leading to an adversarial dynamic. Moreover, the phenomenon of concept drift, where transaction patterns evolve, further complicates model accuracy, often needing an adjustment of the threshold or retraining.

To mitigate the risk of overfitting and ensure realistic performance estimation, forward-validation schemes that preserve the temporal sequence of data are imperative, as they prevent information leakage and bias, thus maintaining the integrity of model evaluation. Many methodologies in literature adhere to these forward-validation principles, offering robust approaches to handle the inherent complexities of FD in dynamic environments.

For instance, fixed rolling origin and cross-validation (rolling window) maintain temporal order. The fixed rolling origin scheme generates forecasts from a single starting point, holding the training period constant, while the rolling window technique rolls or expands the training window over time, incorporating more historical data as the model progresses (Inoue et al., 2017; Tashman, 2000). Additionally, walk-forward validation divides the data into training and testing sets while moving forward in time, ensuring each iteration includes recent data and closely simulates real-world applications (Brownlee, 2018).

2.3.3. Sampling

Random undersampling is a technique used to address the issue of class imbalance in datasets, which is a common problem in ML tasks such as classification and is very acute in FD. The technique involves reducing the number of instances in the majority class, in this case, genuine transactions, to match the number of instances in the minority class – fraudulent transactions - thereby balancing the class distribution (Chawla et al., 2002).

A popular technique for addressing undersampling in a different way is prototype generation, wherein the K-Means algorithm is employed to find data cluster centroids (Triguero et al., 2012). Subsequently, new majority class instances are generated based on these centroids while retaining the minority class instances. However, the quality of the genuine cases depends on the quality of the clusters, which may still be subject to label noise.

Alternatively, some approaches in the literature use distances to eliminate redundant majority class events or to remove noisy and borderline majority class examples (Menardi & Torelli, 2014; Tomek, 1976; Yoon & Kwek, 2005; Zhang & Mani, 2003). These methods focus on refining the dataset by removing instances that do not contribute significantly to the model's learning process.

2.3.4. Supervised Learning

Supervised algorithms are very popular and extensively employed in FD. These methods operate based on the assumption that the used dataset contains labelled instances categorized as either normal or anomalous and requires confidence in the accurate labelling of the original data, in this case, fraudulent or non-fraudulent (Bolton & Hand, 2001; Hilal et al., 2022). Supervised Learning takes advantage of historical transactions by identifying fraudulent patterns (Carcillo et al., 2021).

Despite its advantages, supervised learning has its limitations including inherent rarity of occurrence of deceptive instances (Bolton & Hand, 2002; Hilal et al., 2022). This challenge is known as the class imbalance problem and imposes a problem because ML classifiers trained on imbalanced data tend to exhibit a bias toward overpredicting the majority class, resulting in a high misclassification rate for the minority class (Esposito et al., 2021). Another limitation related to labels is that the process of obtaining labels of the anomalous class can be challenging due to the nature of the problem (Hilal et al., 2022). Both challenges will be addressed later.

Popular techniques in the domain of FD include tree-based models and deep learning (Ali et al., 2022; Grinsztajn et al., 2022).

2.3.4.1. Tree-based models

Decision trees, Random Forests, Gradient Boosting, and other tree-based methods remain prominent in the field due to their simplicity and adaptability in managing several data types, commonly found in financial transactions (Xuan et al., 2018).

Random Forests appear as the most used technique in the FD literature. Notably, it produces superior results in regression tasks by outputting the median prediction and in classification problems by determining the mode of the classes from single trees. This is supported by recent studies affirming the superior performance of Random Forest compared to other methods (Ali et al., 2022; Ashtiani & Raahemi, 2022; Xuan et al., 2018).

Specifically, Xuan et al. (2018) demonstrated the effectiveness of Random Forests in credit card FD through extensive experiments, favouring the classification-and-regression-tree-based approach. Similarly, Grinsztajn et al. (2022) found that tree-based models, including Extreme Gradient Boosting (XGBoost) and Random Forests, outperformed deep learning methods on tabular data, particularly in terms of accuracy and computational efficiency.

2.3.5. Unsupervised Learning

Unsupervised models play a crucial role in the field of FD, particularly in scenarios where labelled instances of fraud are limited or dynamic fraud patterns emerge (Bolton & Hand, 2001; Rezapour Mashhadi, 2019).

In unsupervised methods, such as clustering and compression algorithms, fraudulent instances are viewed as outliers, and so the goal is to identify the instances that are different from the norm (Bolton & Hand, 2002; Carcillo et al., 2021). If this assumption of fraudulent instances being uncommon is not found, it may result in an elevated FPR (Nassif et al., 2021).

When employing supervised algorithms, adapting to changes in customer behaviour and the creation of new fraud patterns by fraudsters pose a significant challenge. In such scenarios, the use of unsupervised learning techniques benefits FDS in identifying irregular patterns (Carcillo et al., 2021).

In some instances, determining if a transaction is fraudulent can be unclear, especially when the system did not complete the transaction. In such cases, unsupervised methods are preferred for analysis (Rezapour Mashhadi, 2019).

Prominent models in the existing literature encompass k-means, SVMs, and Hidden Markov Models (Al-Hashedi & Magalingam, 2021; Rezapour Mashhadi, 2019; Yu et al., 2015).

2.3.6. Ensemble Learning

Ensemble learning is a concept widely explored in the literature, that involves combining predictions from multiple models to enhance performance in classification and regression tasks by harnessing the strengths of several algorithms. It is considered the state-of-the-art technique in many ML problems, including anomaly detection and FD (Dong et al., 2020; Mienye & Sun, 2022; Sagi & Rokach, 2018).

In the context of FD, Ensemble methods are prominent because they provide robust results and show great capacity when dealing with imbalanced data (Mienye & Sun, 2022; Nassif et al., 2021).

In the realm of Ensemble Learning, famous techniques, also employed in FD, are Random Forest, Boosting, and Bootstrap Aggregation (Bagging) along with their variations. Each ensemble method has a specific goal, for instance, Random Forests focus on reducing overfitting, boosting focuses on reducing bias, Bagging on minimizing variance, and stacking on improving overall predictions (Ali et al., 2022; Ashtiani & Raahemi, 2022; Breiman, 2001).

While ensemble techniques are more commonly associated with supervised learning, they have also been proposed and employed in unsupervised and mixed supervised-unsupervised settings. For example, Carcillo et al. (2021) implemented a hybrid approach combining unsupervised outlier scores to augment the feature set of a FD classifier. The experiment found their approach to be efficient, ultimately improving detection accuracy. They state that using both unsupervised and supervised methods is complementary. Supervised methods learn from past fraud behaviour, while unsupervised methods help find new fraud patterns. This is especially helpful when there is a huge amount of data without labels and only a few labelled instances, also starting to come in the scope of semi-supervised learning.

This strategy of combining unsupervised learning and supervised learning was first introduced by Micenková et al. (2014), who named it the principle of *best-of-both-worlds*. Their approach involved sequentially applying various unsupervised outlier detection algorithms to enhance a dataset with outlier scores. This augmented dataset blends the original data with outlier scores, boosting classifier accuracy. The approach efficiently captures diverse feature space outliers without requiring score normalization or heterogeneous method combination. The supervised method then autonomously extracts valuable information from these scores.

2.3.6.1. Light Gradient Boosting Machine

Light Gradient Boosting Machine (LGBM) is a Gradient Boosting Decision Tree framework that employs innovative techniques such as Gradient-based One-Side Sampling and Exclusive Feature Bundling (Friedman, 2001; Ke et al., 2017).

LGBM works by building an ensemble of decision trees, where each tree is trained to correct the errors of its predecessors. During training, it uses Gradient-based One-Side Sampling to prioritize data instances with larger gradients, ensuring the model focuses on more informative samples. Simultaneously, Exclusive Feature Bundling reduces memory consumption and speeds up the training process by combining features that rarely take non-zero values simultaneously. These strategies contribute to LGBM's effectiveness in handling large datasets and improving model performance (Ke et al., 2017).

2.3.6.2. Extreme Gradient Boosting

Extreme Gradient Boosting (XGBoost) is a powerful and efficient implementation of the gradient boosting framework. Developed by Chen & Guestrin (2016), XGBoost is designed to optimize both computational speed and model performance. It achieves this through a variety of advanced features and techniques.

XGBoost uses a boosting method that builds an ensemble of decision trees sequentially, where each tree attempts to correct the errors of the previous ones. It incorporates a regularization term in its objective function to prevent overfitting and improve model generalization. Additionally, XGBoost employs techniques such as shrinkage (learning rate) to scale the contribution of each tree and column subsampling to reduce overfitting and enhance model robustness.

A standout feature of XGBoost is its ability to handle missing values efficiently by automatically learning the best imputation direction. It also supports parallel processing, which significantly speeds up training times by distributing computations across multiple CPU cores. These features, combined with its flexibility to handle various types of data and model configurations, make XGBoost a popular choice for many ML tasks, including classification and regression.

2.3.6.3. Categorical Boosting

Categorical Boosting (CatBoost) stands out as a robust ML library introduced to tackle challenges associated with categorical features (*CatBoost — Yandex Technologies, 2017*). In response to the limitations of standard gradient boosting algorithms, the developers proposed an ordering principle, giving rise to ordered boosting.

This modification prevents target leakage by training each tree with a different random permutation of the dataset. This ensures that the prediction for any data point is made without using its own target value. Additionally, CatBoost converts categorical features into numerical values using a process called target-based statistics, where it calculates and incorporates the average target value for each category. This approach effectively captures the information within categorical features while avoiding overfitting. Combined, these techniques enable CatBoost to efficiently handle datasets with both numerical and categorical variables, enhancing model accuracy and robustness for classification and regression tasks (Prokhorenkova et al., 2018).

The proposed solution is encapsulated in the open-source CatBoost library, offering a powerful solution for both classification and regression tasks.

2.4. META LEARNING

Meta learning, often referred to as "learning to learn", is an advanced ML paradigm designed to enhance the adaptability and efficiency of models by leveraging prior learning experiences. The core idea behind meta-learning is to enable models to quickly adapt to new tasks with minimal data and training, by learning from a diverse set of previous tasks (Hutter et al., 2019). This approach is particularly beneficial in scenarios where data is scarce or expensive to obtain, and quick adaptation is crucial.

Meta learning can be broadly categorized into three main types (Gharoun et al., 2024; Tian et al., 2022):

- **Metric-based:** Metric-based methods focus on learning a similarity measure to compare query samples against a support set, predicting labels based on proximity. The core idea is embedding learning, where high-dimensional samples are converted into lower-dimensional vectors, and similar samples are placed closer together in this space. Techniques such as Siamese Networks, Prototypical Networks, and Matching Networks fall under this category. These methods rely on embedding functions and similarity metrics like Euclidean distance or cosine similarity to make predictions based on nearest neighbor classification in the embedding space.
- **Memory-based or Model-based:** Memory-based methods enhance meta-learning by incorporating memory components to store and retrieve information from previous tasks, enabling better adaptation to new tasks. These methods include architectures like Memory-Augmented Neural Networks, which use external memory for read/write

operations, and models like Conditional Neural Processes (CNPs) which use internal memory to aggregate representations of support sets for efficient predictions. These techniques allow the model to leverage past experiences dynamically, improving the learning process for new tasks.

- **Learning-based or Optimization-based:** Learning-based methods involve training meta-learners to optimize certain aspects of the learning process itself, such as initialization, parameter adjustment, or optimization strategy. Examples include Model-Agnostic Meta-Learning, which learns an optimal initialization for fast adaptation to new tasks, and methods that generate task-specific parameters through a meta-learner. These approaches aim to create flexible, adaptable models that can quickly adjust to new data with minimal additional training, often leveraging gradient-based updates and other optimization techniques to enhance learning efficiency.

Meta learning has been successfully applied in various domains, including computer vision, natural language processing, and RL. In computer vision, meta learning techniques have been used to achieve state-of-the-art performance in few-shot image classification and object detection (Shaha & Pawar, 2018). In natural language processing, meta learning has enabled models to quickly adapt to new languages or domains with limited training data (Ruder et al., 2019).

The integration of meta learning into FDS can hold significant potential. By leveraging meta learning, these systems can quickly adapt to emerging fraud patterns and new types of fraudulent activities (Tian et al., 2022). This adaptability is crucial in financial environments where fraud strategies evolve rapidly.

2.5. REINFORCEMENT LEARNING

Earlier, supervised learning and unsupervised learning were discussed; here, the third branch of ML, RL is presented. The main idea in RL is that the learning process for solving a decision-making problem involves a sequence of trial and error. In specific, an agent learns in an interactive environment by trial and error using feedback from its actions and experiences.

Unlike supervised learning where a model is trained with a labelled dataset, RL uses a reward system to guide the learning process, aiming to maximize cumulative rewards over time. This is achieved through the agent's decisions, which are refined progressively as it learns to map environmental states to the optimal actions based on reward feedback. Key components include the definition of an environment, a set of states, actions, and rewards which can vary over time and with different policies (Sutton & Barto, 2018).

In the RL framework, the agent interacts with the environment by taking actions based on its observations, forming a trajectory of states and actions. The environment responds to each action with a reward and a new state, enabling the agent to learn which actions yield the highest rewards over time. The reward is a scalar signal indicating the immediate benefit of an action, but its true value may be delayed, reflecting long-term outcomes. The ultimate goal

is to maximize the cumulative reward, known as the return, which is calculated as the total discounted reward from a given point in time. The discounted reward is the total cumulative reward an agent aims to maximize, where future rewards are multiplied by a discount factor to reflect their present value and ensure the sum converges to a finite value.

A crucial aspect of RL is the concept of the Markov state, which encompasses all the information necessary to make future decisions, independent of past states. This property allows the agent to base its actions solely on the current state without needing to consider the entire history. The distinction between fully and partially observable environments is significant: in fully observable environments, the agent has complete access to the state, whereas, in partially observable environments, the agent only receives partial information. Understanding these concepts is fundamental to developing effective RL algorithms that can learn and adapt in complex, dynamic environments.

In FD, RL can be particularly effective due to its adaptability and ability to continually learn and adjust to new patterns of fraudulent activity. Traditional ML models often require retraining with new data to remain effective, but a RL system can theoretically adjust in real-time, learning from each new transaction whether its previous actions (e.g., flagging a transaction as fraudulent) were successful (Mead et al., 2018; Vimal et al., 2021).

While RL offers considerable advantages, it also comes with unique challenges. The complexity of RL models can make them difficult to implement and maintain. Additionally, they require continuous interaction with the environment, which can be resource-intensive. Designing an appropriate reward system that accurately reflects the trade-offs between detecting more fraud and reducing false positives is also challenging. These challenges must be carefully considered to ensure the successful deployment of RL in FDSs.

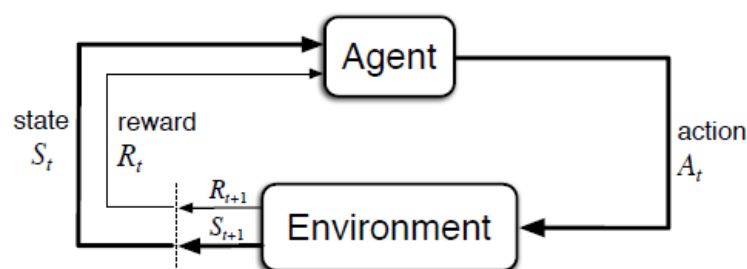


Figure 5 – Agent-Environment Interaction Flow (Sutton & Barto, 2018)

2.5.1. Markov Decision Process

The Markov Decision Process (MDP) is a formal framework used to model decision-making scenarios, particularly useful in optimisation and dynamic programming. The primary objective of an MDP is to determine the optimal action to take in each state to maximize the expected cumulative reward over time. This framework provides a structured method for

solving problems that involve making sequential decisions, each affecting future states and rewards, and is fundamental to the application of RL algorithms.

An MDP is defined by the 5-tuple (S, A, P, R, γ) . Where:

- S is a finite set of states
- A is a finite set of actions
- P is a state transition probability matrix, defined as follows:

$$P_a(s, s') = P(s_{t+1} = s' \mid s_t = s, a_t = a) \quad (2)$$

- R is a reward function, defined as follows:

$$R_a(s) = E[r_{t+1} \mid s_t = s, a_t = a] \quad (3)$$

- γ is a discount factor, $\gamma \in [0, 1]$

2.5.2. Bellman Equation

The Bellman equation provides a recursive decomposition for solving MDPs by expressing the value of a state as the expected return of the best action taken from that state. It helps in determining the optimal policy by breaking down the value function into the immediate reward plus the discounted value of the next state. The Bellman equation is formulated as follows:

$$V(s) = \max_a (R(s, a) + \gamma \sum_{s'} P(s'|s, a) V(s')) \quad (4)$$

where $V(s)$ is the value of s , $R(s, a)$ is the reward for taking action a in state s , γ is the discount factor, $P(s'|s, a)$ is the transition probability, and $V(s')$ is the value of the next state s' .

2.5.3. Types of RL Algorithms

In RL, there exist model-free and model-based approaches. Model-free methods do not require knowledge of the environment's dynamics, learning directly from interactions with the environment to optimize policies. Conversely, model-based methods involve building a model of the environment's dynamics and using this model to plan and make decisions, with examples including Value Iteration and Policy Iteration.

RL can also be categorized into value-based and policy-based methods. Value-based approaches focus on learning the value function, which estimates the expected return of states or state-action pairs, and then derive policies from these value functions, with Q-Learning being a prominent example. Policy-based methods directly learn the policy that maps states to actions without explicitly estimating value functions, often optimizing the policy using gradient-based methods.

2.5.4. Q-Learning

Q-Learning is a popular RL algorithm used to find the optimal policy for a given finite MDP. It is a model-free algorithm, and the core idea of Q-Learning is to learn a value function, $Q(s, a)$ which represents the expected utility of taking action a in state s and following the optimal policy thereafter (Francois-Lavet et al., 2018).

In Q-Learning, an agent interacts with the environment in discrete time steps. At each step, the agent observes the current state s chooses an action a based on a policy, receives a reward r , and transitions to a new state s' . The value function $Q(s, a)$ is iteratively updated using the Bellman equation to approximate the maximum expected future rewards.

The update rule for Q-Learning is given by:

$$Q(s, a) \rightarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)] \quad (5)$$

Where:

- α is the learning rate, $\alpha \in]0,1]$
- γ is the discount factor, $\gamma \in [0,1]$
- r is the immediate reward received after taking action a
- $\max_{a'} Q(s', a')$ is the maximum expected future reward for the next state s'

Q-Learning is particularly effective because it approximates the optimal action-value function independently of the policy being followed, simplifying the analysis and convergence of the algorithm. If all state-action pairs are sufficiently explored, Q-Learning is guaranteed to converge to the optimal Q-values. This makes Q-Learning a powerful method for RL tasks where the agent seeks to learn optimal behaviours through interaction with its environment. The pseudocode of this algorithm can be found in the Annex.

2.5.5. Deep Q-Learning

Deep Q-Learning is an extension of the traditional Q-Learning algorithm introduced by Mnih et al. (2015) to handle environments with large or continuous state spaces. Instead of using a Q-table to store state-action values, Deep Q-Learning employs a deep neural network to approximate the Q-function. This neural network takes the state as input and outputs Q-values for all possible actions, enabling the agent to make decisions even in high-dimensional spaces.

The primary innovation in Deep Q-Learning is the use of experience replay and target networks to stabilize training. Experience replay involves storing the agent's experiences (state, action, reward, next state) in a buffer and sampling random mini-batches from this buffer to train the network. This technique breaks the correlation between consecutive experiences, leading to more stable learning.

Additionally, a target network, which is a copy of the Q-network, is updated periodically and used to compute the target Q-values. This helps to prevent large oscillations in the Q-values during training. The main update rule in Deep Q-Learning, leveraging experience replay and target networks, is given by:

$$Q(s, a) \rightarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q_{target}(s', a') - Q(s, a)] \quad (6)$$

Where Q_{target} is the target network used to compute the target Q-values.

2.5.6. Double Deep Q-Learning

Double Deep Q-Learning addresses a specific limitation of Deep Q-Learning, which is the overestimation bias of action values (Francois-Lavet et al., 2018). This bias occurs because the same network is used to select and evaluate actions, leading to an upward bias in the estimated Q-values.

A Double Deep Q- Network (DDQN) mitigates this issue by decoupling the action selection and action evaluation processes. It uses the Q-network to select the best action for the next state and the target network to evaluate this action. This results in more accurate Q-value estimates and improved performance. The update rule for DDQN is given by:

$$Q(s, a) \rightarrow Q(s, a) + \alpha[r + \gamma Q_{target}(s', \arg \max_{a'} Q(s', a')) - Q(s, a)] \quad (7)$$

In 7, the action $\arg \max_{a'} Q(s', a')$ is selected using the Q-network, while the value is estimated using the target network. This separation helps in reducing the overestimation bias and leads to more reliable learning outcomes.

Both Deep Q-Learning and Double Deep Q-Learning represent significant advancements in RL, allowing agents to learn effectively in complex environments with high-dimensional state spaces.

2.6. CHALLENGES OF FRAUD DETECTION

2.6.1. Imbalance Data

When it comes to FD using ML, the main challenge is the imbalance in the dataset (Behdad et al., 2012; Gao et al., 2020). This occurs when instances of a certain class far outnumber the other, posing a challenge in building effective models. In FD, the minority class, representing fraudulent transactions, is particularly susceptible to misclassification (Yu et al., 2015).

The problem arises because ML classifiers tend to favour the majority class during training, leading to biased predictions. This imbalance results in a higher misclassification rate for the minority class, compromising the model's ability to accurately identify fraud (Esposito et al., 2021).

To tackle this, several solutions have been proposed. These solutions can be categorized into four main groups: algorithm level, sampling strategy, cost-sensitive learning and ensemble based and hybrid methods.

In algorithm level, the classifier learning procedure can be altered through various methods. Examples include:

- Adjusting the splitting criteria in decision tree induction (Boonchuay et al., 2017; Cieslak & Chawla, 2008);
- Implementing weighted approaches to assign greater importance to specific training instances to boost their influence on the algorithm prediction (Yang et al., 2007);
- Employing active learning during training to select the most informative instances and enhance the quality of the resulting classifier (Aggarwal et al., 2020; Fu & Lee, 2013; Zhang et al., 2014).

In terms of sampling strategy, a traditional approach for addressing imbalanced data involves random undersampling, which aims to balance the minority class by randomly removing instances from the majority class. However, this indiscriminate method may lead to the discarding of potentially valuable data, complicating the learning of the decision boundary between minority and majority instances and resulting in a decline in classification performance. Additionally, considering label noise and potential delays in the fraud labeling process, it becomes essential to accurately filter certain events from the majority class that do not contribute to the algorithm's ability to assign correct risk scoring. Consequently, several state-of-the-art alternatives have been proposed to overcome this limitation.

Among them, there are resampling methods such as oversampling, undersampling, and hybrid sampling aim to balance class distribution.

Cost-sensitive learning that assigns different misclassification costs to classes, prioritizing the minority class; Decision threshold adjustment which involves fine-tuning classification thresholds, offering a nuanced approach; Ensemble learning that, as mentioned previously, consists of combining outputs from multiple models, providing a robust solution. Furthermore, it is also important to use the proper evaluation metrics and hyperparameters (Behdad et al., 2012; Yu et al., 2015).

These diverse approaches underscore the ongoing efforts to enhance the effectiveness of ML models in FD, ensuring a multifaceted and adaptive response to class imbalance challenges.

2.6.2. Label Noise

Using recent data (near-real-time data) poses a significant challenge of having noise in the data labels. This noise can happen due to the presence of a cold start in the labels, delays in the fraud labelling process, misclassification cases originated from analysts, the system's

operational strategy or call centers where the individual contacted is not the victim but the fraudster.

The cold start problem is essentially an absence in the data, in this case, the target variable, at the beginning and there is a delay in obtaining it, due to the process of collecting the label being slow and costly (Barata et al., 2021).

To solve the cold start problem, unsupervised learning algorithms can be used once they don't require labelled instances. However, active learning has also emerged as a solution. Active Learning operates by strategically selecting observations from an unlabelled set for manual labelling, aiming to improve the classifier's performance. Instead of passively relying on pre-labelled data, active learning actively engages by identifying instances where the model is uncertain or likely to benefit from additional information (Barata et al., 2021; Carcillo et al., 2018).

The misclassified instances can appear if, for instance, the system opts to block suspicious transactions rather than suspending the associated account or card. There is still a possibility that victims may overlook fraudulent activities, leading to inaccurate feedback for case management by responsible institutions.

2.6.3. Interpretability

Despite ML models being efficient in generating accurate predictions, minimising costs, and improving customer satisfaction, the lack of interpretability in decision-making poses a critical concern. Many of these models operate as black boxes, where the only information known is the inputs and outputs but not the internal process (Chaquet-Ulledemolins et al., 2022a).

The interpretability issue becomes more pronounced when addressing fraud occurrences, where traceability is fundamental to understanding what triggered the occurrence of being flagged as fraudulent (Rodrigues et al., 2022).

2.7. RELATED WORK

Dynamic optimisation is not a common practice in ML, particularly for addressing the ever-changing nature of real-world data. Furthermore, in the field of FD, literature is sparse due to the confidentiality maintained by companies handling sensitive data.

Nevertheless, Gao et al. (2020) address the challenge of optimising decision thresholds in the context of classifying imbalanced multi-label data. Their study introduces the Adaptive Decision Threshold-Based Extreme Learning Machine (ADT-ELM), an enhancement of the Extreme Learning Machine framework. The key innovation in ADT-ELM is the use of adaptive decision thresholds, which are optimized using Particle Swarm Optimization to maximize the F-measure metrics. This adaptive approach ensures that the classification model can effectively handle imbalanced data by adjusting thresholds to improve overall performance. The research demonstrates the effectiveness of ADT-ELM on twelve benchmark datasets,

showing significant improvements in classification accuracy and computational efficiency compared to other multi-label learning algorithms. The adaptive threshold optimization proves crucial in managing class imbalances, highlighting its importance for improving model performance in complex datasets.

In the field of financial activities, Herasymovych Märka and Lukason (2019) explored the dynamic optimisation of acceptance thresholds in credit scoring models using RL. Traditional credit scoring methods often utilize static thresholds, which can become suboptimal as data distributions and business objectives evolve. The study proposes a Q-learning-based approach to continuously adjust the acceptance threshold in real time, aiming to optimize business outcomes such as profit and risk management. This RL-based method allows the model to adaptively find the optimal threshold by learning from ongoing interactions with the environment, thereby improving upon traditional cost-sensitive optimization techniques. The results show that the RL approach effectively balances the trade-off between minimizing default rates and maximizing acceptance rates, providing a flexible and adaptive solution for credit scoring.

In their study, Vimal et al. (2021) explore the application of deep RL to payment FD, presenting a novel approach to address the limitations of traditional supervised learning methods. These methods often struggle with high-class imbalances, dynamic fraud patterns, and delayed feedback. By framing FD as a sequential decision-making problem, Vimal et al. utilize Deep Q-learning to balance FD rates with transaction approval rates effectively. Their work emphasizes the adaptability of deep RL in responding to changing fraud patterns, demonstrating its potential to improve decision-making processes in real-time. This approach highlights a promising direction for future advancements in the field of FD.

Moreover, the article "Deep Reinforcement Learning for Imbalanced Classification" by Enlu Lin, Qiong Chen, and Xiaoming Qi (2019) also addresses the challenge of class imbalance in ML. The authors propose a novel model that frames the imbalanced classification problem as a sequential decision-making process and uses a Deep Q-Learning Network (DQN) to develop an optimal classification policy. The agent in this model classifies samples and receives rewards that are adjusted to make it more sensitive to minority class samples. Through this approach, the model demonstrates improved performance in identifying minority samples compared to traditional methods. The study highlights the potential of deep RL in handling imbalanced data, offering a promising solution to enhance classification accuracy in various real-world applications.

These studies emphasize the critical role of dynamic techniques in enhancing the performance of FDS. By focusing on adaptive ML and RL techniques, they contribute valuable insights into how dynamic approaches can effectively address challenges in imbalanced data classification and financial risk management.

3. METHODOLOGY

The Cross Industry Standard Process for Data Mining (CRISP-DM) Methodology has long been the reference for ML practices. However, the landscape of this field is evolving, problems are becoming more complex, and projects are moving beyond traditional frameworks.

The CRISP-DM steps include Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, and ends with the Deployment (Chapman et al., 2000). This framework assumes that the data context does not change, which is not true for many real-world problems (Martínez-Plumed et al., 2017).

The fraud patterns are always changing as the fraudsters are always evolving their techniques and so the static assumption of CRISP-DM does not hold. Martínez-Plumed et al. (2017) proposed an extension of CRISP-DM, called Context Aware Standard Process for Data Mining (CASP-DM), that addresses challenges in managing context and model reusability. Furthermore, they state that a proactive approach to managing context changes involves building a versatile model not tailored to specific contexts or changes, facilitating model reusability.

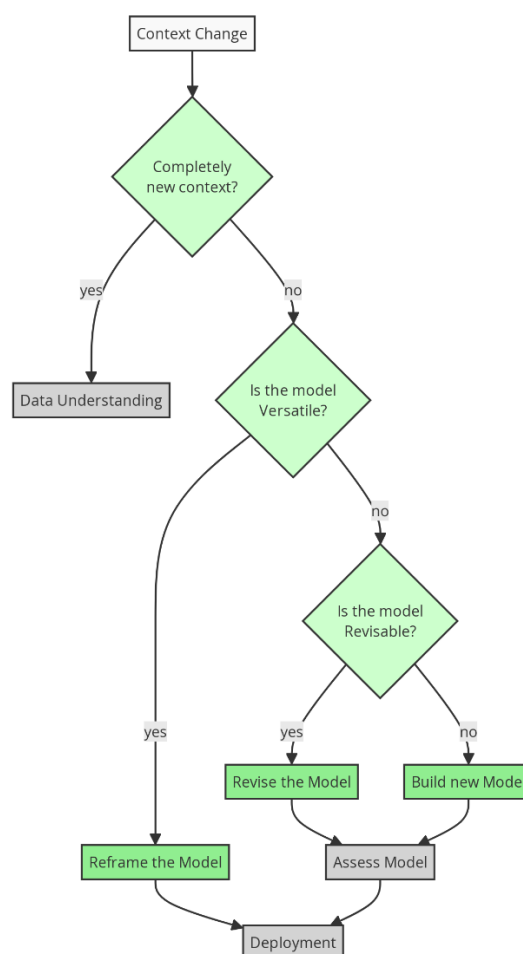


Figure 6 - Simplified CASP-DM Diagram

In Figure 6, it is possible to see a simplified diagram of the CASP-DM methodology where the dark green boxes lead back to CRISP-DM. Whenever a new context arises, such as a new type of fraud or a data drift, CASP-DM guides on when to re-apply CRISP-DM, avoiding starting the process from scratch.

To make the model versatile, emphasis was placed on implementing dynamic techniques to ensure adaptability to varying conditions and improve overall performance.

For this purpose, several strategies were explored. Initially, the focus was to build an adaptative threshold, this involved employing a secondary model to refine the threshold of the baseline model. Building on this, RL techniques were also explored. This approach uses a reward-based system where the RL agent learns to make real-time decisions based on feedback from its performance, eliminating the need for predefined risk scores and thresholds.

The application of these dynamic methods is detailed below, following a description of the necessary preliminary steps, which were crucial for developing the strategies effectively. The process begins with identifying the problem and understanding the business context. This is followed by exploring the available data, creating new features, defining the sampling strategy and finally, developing the proposed strategies.

3.1. PROBLEM IDENTIFICATION

The first step was to understand the problem thoroughly to identify the most suitable solution. As mentioned in the introduction, the challenge to be tackled is striking a balance between achieving a service acceptance rate that aligns with the high-quality standard of user experience and effectively detecting fraud (Alharbi et al., 2022; Mehbodniya et al., 2021). In technical terms, the objective established to address this problem was to ensure that the FPR remains below a predetermined threshold while maximizing the TPR.

3.2. DATA EXPLORATION

In this stage, a comprehensive dataset evaluation is conducted to gain familiarity with its contents. This involved several key activities such as examining the fraud rate, assessing data quality, exploring feature correlations, identifying new features, addressing data cleaning requirements, and obtaining other valuable insights for the next steps.

The dataset used for this study is a real-world transaction dataset provided by a major payment processor in Europe. It was stored in the Hadoop Distributed File System and, therefore, Spark was used to efficiently handle and process the data.

The dataset comprises 1.4 million instances, each corresponding to a single transaction with 35 natural features. Spanning 10 months, the dataset exhibits a fraud rate of 0.2%, highlighting the significant class imbalance.

The period of these datasets was carefully assessed. This aspect is critical in FD projects due to data drift. Fraud patterns can change over time and understanding these shifts is crucial for selecting appropriate training and testing periods, which will be further elaborated in the Data Split Section.

The dataset includes a “DateTime” variable, providing detailed temporal information about the transactions down to the second. The dataset encompasses data from a span of 10 months, allowing for precise analysis of temporal patterns and trends over this period.

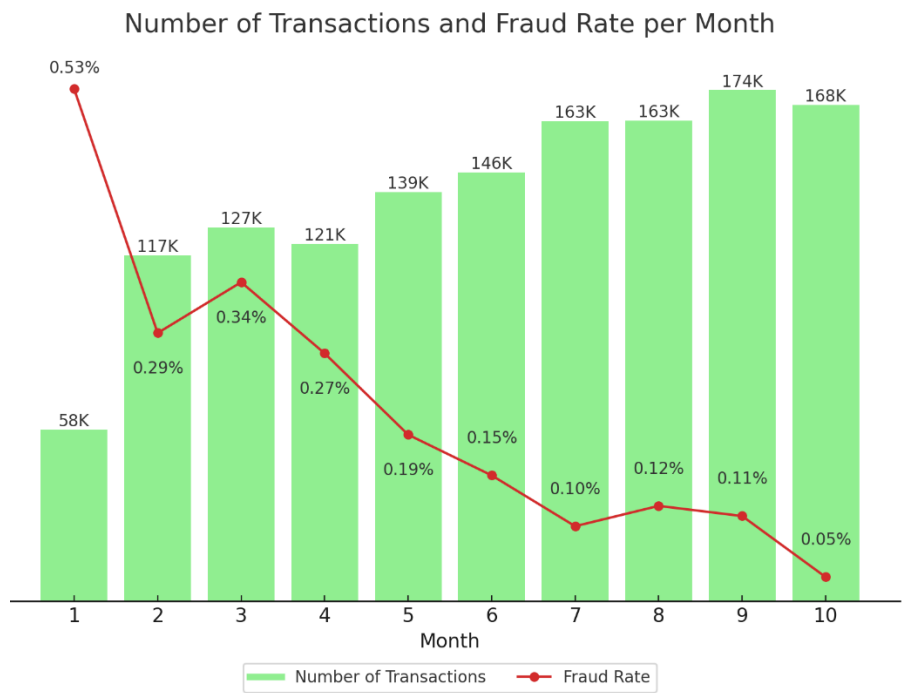


Figure 7 - Monthly Transaction Volume and Fraud Rate

As observed, the number of transactions has been gradually increasing, possibly indicating the digitization of payments, which underscores the need for an effective FDS. The fraud rate has progressively decreased each month, reaching a low of 0.05% in the last month. This decline does not necessarily imply a reduction in fraud incidents but may reflect the efficiency of the FDS in blocking fraudulent transactions, which can result in a lack of feedback on the true labels—a problem discussed previously.

During this exploration, it was noticed that the transactions were divided into four different use case segments, these segments have been categorised as A, B, C and D for anonymisation reasons. From a financial point of view, these segments were found interesting and showed different behaviours with the label. The fraud rates observed for each segment indicate this difference, as illustrated in the following figure and table.

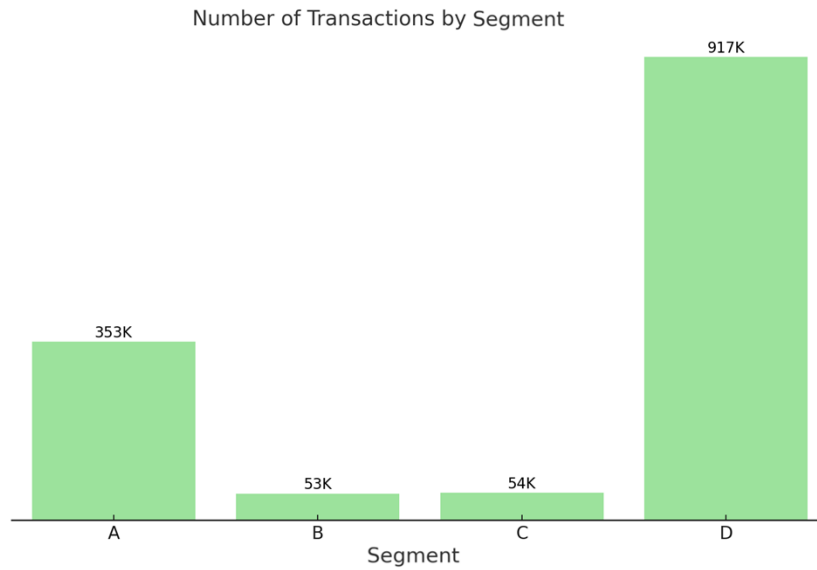


Figure 8 - Transaction Volume by Segment

Table 1 - Fraud Rate by Segment

Segment	Fraud Rate
A	0,06%
B	1,59%
C	0,40%
D	0,14%

From the above analyses, it's evident that Segment D processes the most transactions with more than 900 thousand transactions in the 10-month period, yet it maintains a low fraud rate. Segment B, while handling fewer transactions, experiences the highest rate of fraudulent activities, this being 1,59%, more than twice as large as segment C and more than ten times larger than segments A and D. This discrepancy highlights the importance of segment specific strategies in FD, as the risk levels vary substantially across different use case segments. The implementation of targeted FD models for each segment could enhance the overall efficiency and accuracy of the FDS.

Another important aspect to consider is the volume of transactions. For this analysis, the transaction values were standardized using Min-Max Scaling and then multiplied by ten thousand to facilitate the analysis.

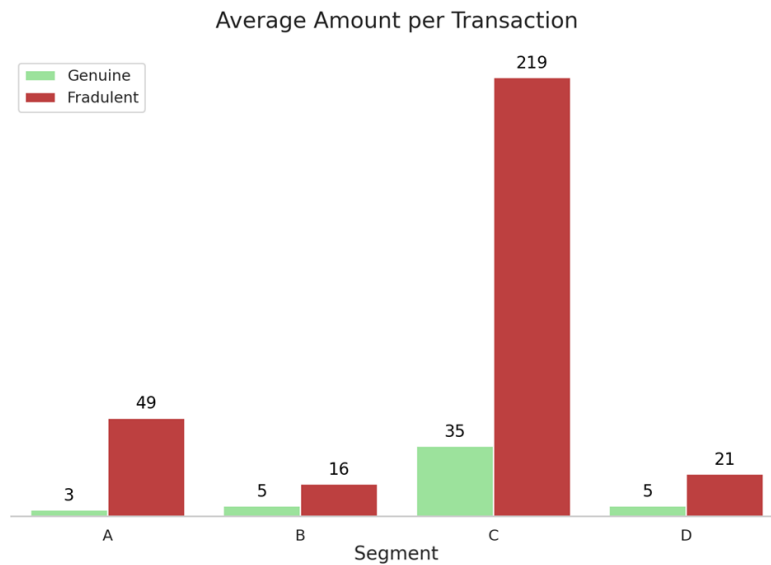


Figure 9 - Average Transaction Amount by Segment

From Figure 9, it is possible to conclude that fraudulent transactions generally have higher average amounts compared to genuine transactions across all segments. This discrepancy is greater in the C segment in which fraudulent transactions have an average amount 6 times greater than the average amount of genuine transactions.

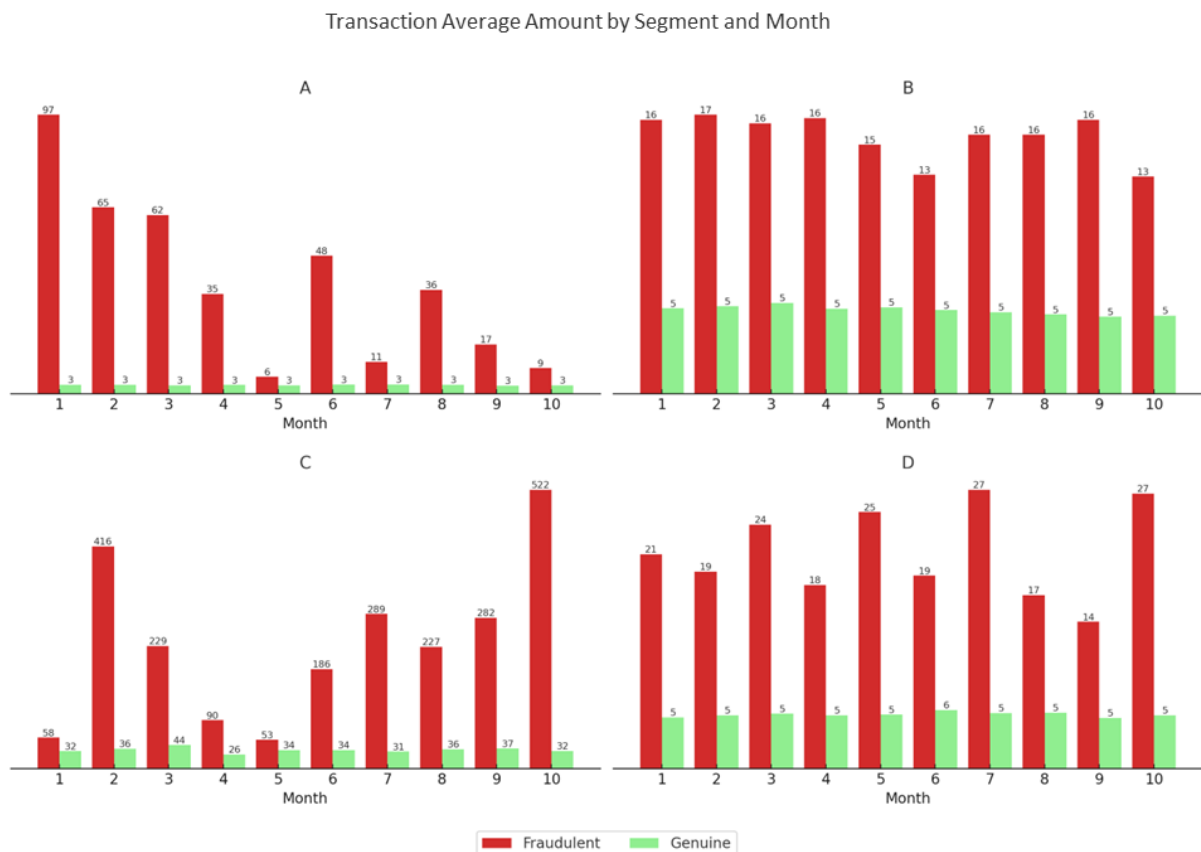


Figure 10 - Monthly Average Transaction Amount by Segment

Figure 10 illustrates the presence of data drift in the dataset. For segments A and C, the average transaction amount shows significant variability over the months. In segment A, for instance, the average amount of genuine transactions remains constant over the 10 months. In contrast, fraudulent transactions gradually decrease until the fifth month, then increase in the sixth month, decrease again in the seventh month, rise in the eighth month, and gradually decrease until the final month.

3.3. FEATURE ENGINEERING

During this stage, additional features were engineered from the original dataset and some transformations were applied to improve the effectiveness of the ML techniques.

A transaction inherently has 35 natural variables. By leveraging these natural variables and integrating data from other information systems, approximately 100 features were created.

Despite their significance in the study, providing a detailed description of each variable would be exhaustive and not add substantial value to the report. These variables encompass several dimensions, including:

- **Customer Behaviour:** Analysing patterns in how customers interact with their accounts and perform transactions.
- **Temporal Data:** Considering the timing of transactions, such as time of day, day of the week, and seasonal patterns.
- **Geographical Data:** Incorporating the location of transactions to detect anomalies based on the geographical context.
- **Comparative Metrics:** Comparing individual transaction patterns against broader population trends to identify outliers.

This comprehensive set of features, developed with input from business experts, ensured the capture of key elements essential for accurate FD. The newly designed features capture the complexities of transaction dynamics, providing a robust foundation for the ML techniques explored in this study.

3.4. SAMPLING

As mentioned in the Literature Review, class imbalance is a common challenge in FD, typically addressed by manipulating the sample through techniques such as undersampling or oversampling. Sampling was a crucial aspect of the entire development of the techniques, warranting special attention. Therefore, this section is dedicated exclusively to the sampling strategies explored throughout the project.

3.4.1. Random Undersampling

Traditional methods often rely on randomness to select genuine instances for removal or fraudulent instances for duplication. One strategy tested in this study was random

undersampling. However, this blind approach can discard potentially useful data, making it harder for the model to learn the decision boundary between minority and majority instances, ultimately diminishing classification performance. Additionally, label noise and delays in the fraud labelling process make removing non-informative events from the majority class essential to enhance the algorithm's results.

3.4.2. Anomaly-Based Undersampling

Inspired by the techniques of prototype generation and alternative distance-based approaches, this study presents an anomaly-based approach that aims to optimize the sample by strategically selecting the observations to remove from the majority class. This approach works with the concept of anomalous instances, which are instances that differ from the rest but are not necessarily fraudulent.

On one hand, genuine anomalous instances can confuse the model if they are similar to fraudulent ones, this can happen because, as previously discussed, a FDS may encounter label noise arising from several factors. On the other hand, if these genuine anomalous instances are dissimilar to fraudulent ones, in this case, they act as outliers, not adding value to the model for learning genuine patterns and so must be removed.

The iForest and its variant, the EIF, were employed to detect anomalous instances as they are effective in identifying instances that deviate from the norm without relying on labels (Liu et al., 2008).

To perform these algorithms, feature selection was performed manually through business knowledge. Note that, for this strategy, the primary goal in selecting features based on business insights was to balance the algorithm's ability to detect genuine anomalous instances rather than achieve perfect results when compared to the labels.

This is crucial because in FD, particularly using unsupervised methods, perfect optimisation could lead to ignoring genuine instances that superficially resemble fraudulent ones. By not solely relying on algorithmic selection of features, we aimed to better capture these subtle nuances, thus supporting the broader objective of enhancing the performance of subsequent steps/techniques.

The choice of hyperparameters such as the number of trees and subsample size was guided by a comprehensive search using a Bayesian optimization approach (Frazier, 2018).

Bayesian hyperparameter optimization uses probabilistic models to search for the optimal combination of hyperparameters, dynamically adapting the hyperparameters based on past evaluations. This approach was used to strike a balance between computational efficiency and a thorough exploration of the hyperparameter space.

The deployment of the model was done using the H2O package due to its compatibility with Spark, facilitating the handling of large volumes of data encountered in this study (H2O, 2024).

3.4.3. Combined Sampling Strategy

A hybrid sampling strategy was implemented to further refine the sample, combining the anomaly-based undersampling with random undersampling. This strategy involves two key steps:

1. Remove Anomalous Instances: Identify and remove the topmost anomalous genuine instances based on their anomaly scores (the top 5%, 10%, 15%, 20% or 30%).
2. Apply random undersampling to the remaining genuine instances to achieve the targeted proportions of fraudulent events (5%, 10%, and 15%).

After deploying iForest and EIF, Spark was no longer used. The data processing needs were sufficiently met by using Pandas and other relevant python packages, which provided simpler means to apply the subsequent techniques.

3.5. THRESHOLD OPTIMIZATION

In this section, the various strategies explored for threshold optimisation are described in detail. These strategies aim to dynamically adjust the decision thresholds of the ML models to enhance FD performance. Each approach, including its methodology and implementation, is thoroughly examined to provide a comprehensive understanding of the optimisation process.

3.5.1. Three Sigma Rule

The first method assessed for setting dynamic thresholds was the 3 Sigma rule, a statistical approach commonly used in anomaly detection. This method operates under the assumption that the data follows a normal distribution. According to the properties of a normal distribution:

- 68% of the data lies within one standard deviation (σ) of the mean (μ).
- 95% of the data falls within two standard deviations.
- 99.7% of the data falls within three standard deviations.

The threshold is defined as:

$$Threshold = \mu + 3\sigma \quad (8)$$

To implement this dynamically, rolling mean and standard deviation can be calculated. This technique specifically relies only on the predicted probabilities (scores), without considering other variables. However, upon further analysis, it was observed that the label scores did not follow a normal distribution, invalidating the assumption of normality. As a result, this method was not pursued further in this study.

3.5.2. Secondary Model

This strategy involves training a secondary ML model with a shorter period of time to identify emerging fraud patterns that are not being captured by the baseline model.

The second model is dynamic, being trained automatically each day. Consequently, certain decisions—such as the choice of model, the method for selecting variables, and the training period—had to be made in advance. These factors are not evaluated daily, as the goal is to create an automated strategy and integrate it into the FDS. Manually analysing these aspects for daily models would be impractical.

This short model integrates with the baseline model by dynamizing the threshold according to the proposed strategy. The defined threshold for each day is the one with an FPR below the target of 0.02, aiming to achieve the maximum possible TPR. A FPR of 0.02 was chosen specifically for the scope of this project and is not based on existing literature or the company's metrics.

3.5.2.1. Model Construction

The process of constructing a secondary model for FD is similar to the standard, found in most ML projects. It involves the selection and evaluation of various ML algorithms, the application of different sampling strategies, the employment of feature selection techniques, and the optimisation of hyperparameters to ensure robust and effective models. The following figure illustrates the steps of the model construction for each day.

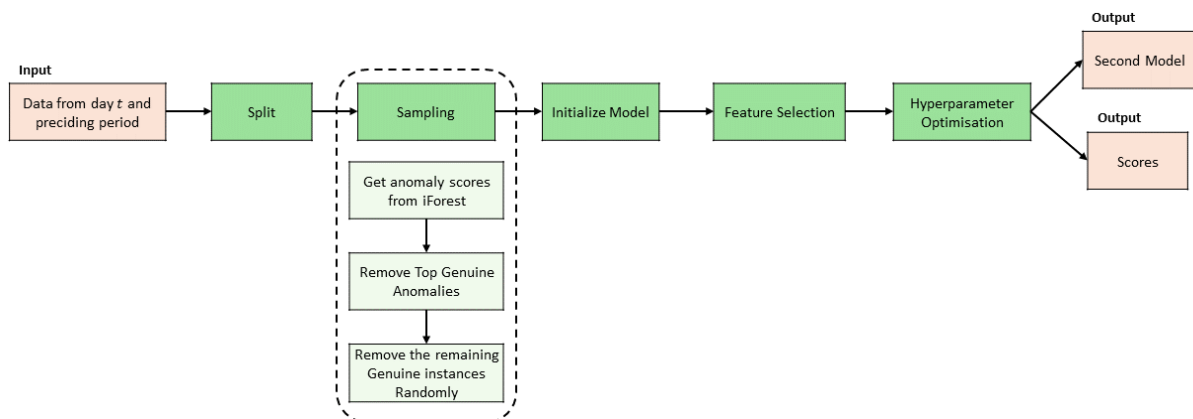


Figure 11 - Second Model Construction

The algorithms assessed in this part of the study were **LGBM**, **XGB**, and **CatBoost**. They were selected due to their demonstrated efficiency in the literature and their proven effectiveness with this type of data and evidenced by business knowledge. These models are known for their strong performance in handling complex datasets and have been widely recognised for their robustness and accuracy in various ML tasks, making them ideal choices for developing robust FD algorithms.

Due to the nature of this strategy, a **walk-forward validation** approach was implemented. The model trained on a preceding period (train set) and tested on a subsequent period (test set), then predicted scores for the target day (validation set). Each day, the model was rebuilt with all periods shifted forward by one day. This ensured continuous updates and accurate performance assessment. Training and testing periods were tested with intervals of one week, two weeks, and one month before the prediction day. Periods exceeding one month were excluded to maintain focus on recent fraud patterns.

The training sets were undersampled to tackle data imbalance. The strategy used was a **combination** of random sampling and anomaly based.

Four methods of feature selection were also performed to identify the most relevant features from the dataset. These methods aimed to select features that would enhance the performance and accuracy of the ML models. The following approaches were explored:

- **Recursive Feature Elimination (RFE):** RFE was applied to iteratively remove the least important features and build a model using the remaining features (X. Chen & Jeong, 2007). This method utilised a LGBM classifier, assessing model performance at each step to determine the optimal subset of features.
- **Feature Importance (FI):** A LGBM classifier was used to calculate the feature's impact on the model's predictions, considering feature dependencies and correlations (Zien et al., 2009). Features with the highest importance scores were selected, ensuring that only the most impactful features were retained in the model.
- **Permutation Importance (PI):** Permutation importance addresses and corrects bias in feature importance (Altmann et al., 2010). It was employed to measure the change in model performance when the value of a feature is randomly shuffled. This technique helped identify features that significantly influenced the model's predictions by observing the decrease in model accuracy when a feature's values were permuted.
- **SHapley Additive exPlanations (SHAP):** SHAP values were used to understand the contribution of each feature to the model's output. By providing detailed insights into how much each feature impacts the prediction, SHAP values enabled the selection of the most influential features, enhancing the model's interpretability and performance (Lundberg & Lee, 2017).

Finally, to improve the model, the hyperparameters were optimized. Similar to the process used in iForest and EIF, **Bayesian hyperparameter optimisation** was employed to enhance model performance.

3.5.2.2. Integration Strategy

Integrating the secondary model into the primary model to dynamically optimize the threshold was a crucial aspect of the research.

Direct Threshold Application

The initial experimentation used a secondary model to dynamically adjust the threshold of the baseline model. This approach aimed to refine the threshold daily based on the secondary model's output. Each day, the secondary model was trained, and the optimized threshold obtained was implemented in the baseline model for the following day.

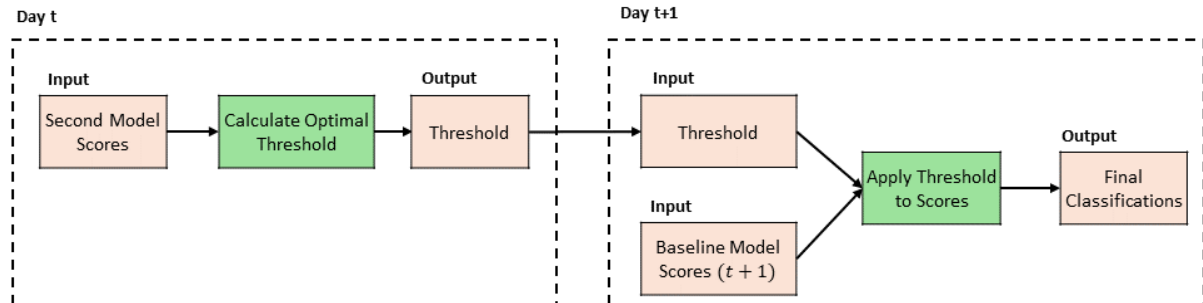


Figure 12 - Direct Threshold Application Strategy

New transactions each day pass through the baseline model and are classified based on the score and the threshold defined by the secondary model, which was set on the previous day. This means that, for final classification, an instance does not go through the secondary model.

Proportional Threshold Adjustment

This approach refines the previous method by indicating how the threshold should be adjusted rather than directly applying the secondary model's threshold to the baseline model. If the threshold of the secondary model increases, the baseline model's threshold also increases proportionally.

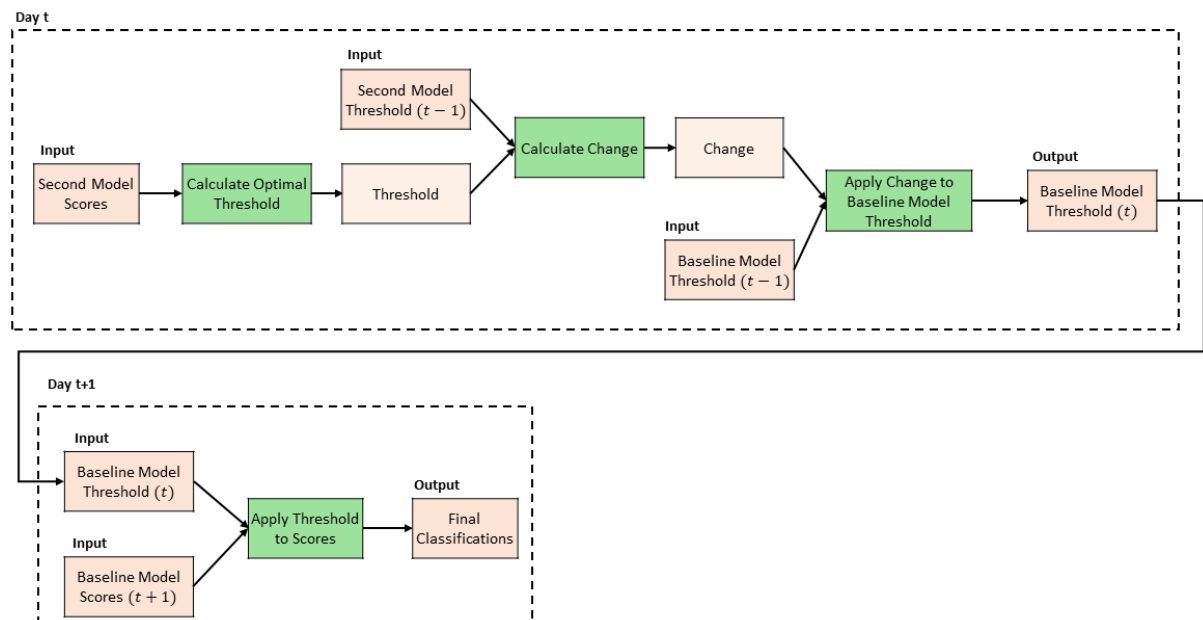


Figure 13 - Proportional Threshold Adjustment Strategy

For instance, if the threshold of the secondary model increases by 5% from one day to the next, the baseline model's threshold is also adjusted by 5%. This approach entails defining an initial threshold for both models to provide a basis for comparison on the first day of the study.

Meta-Learning

Building on the initial approach, a meta-learning strategy was implemented. This strategy involves using a secondary model not just to set a threshold, but to dynamically learn and adjust the threshold based on evolving data patterns.

The meta-model was designed to take the output scores from both the baseline model and the secondary model as inputs. The chosen model was LGBM, selected for its simplicity in this context, as it only requires a low number of input variables. Its hyperparameters were optimized using a Bayesian approach and trained to predict a new, third score.

Based on the results of the meta-model, the threshold for that day was set. This approach requires the validation data to pass through both the second model and the meta-model, unlike the previous approaches where the validation data only went through the baseline model and predicted according to the threshold defined by the second model.

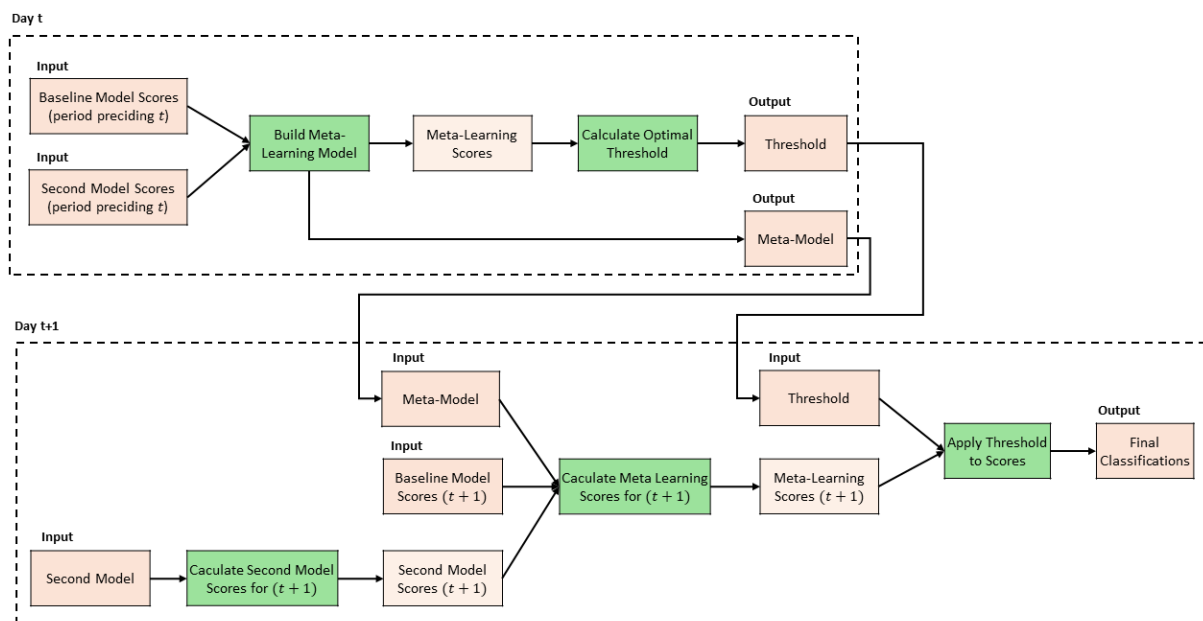


Figure 14 - Meta-Learning Strategy

An intriguing aspect of this approach is that the secondary model has a greater influence on the final classification, as it contributes to both the construction of the score and the threshold, rather than just the threshold as in previous approaches.

3.5.2.3. Segment-Specific Strategy

Recognising that fraud patterns can vary significantly across different contexts, the experimentation further evolved to include segment-specific secondary models. During the

data exploration phase, it was concluded that transactions could be classified into four distinct segments, each exhibiting unique behaviours concerning the label. As a result, when constructing the secondary model, an approach was implemented to create four separate secondary models, one for each segment, rather than a single secondary model.

This segment-specific strategy can be applied using any of the integration strategies previously described. The primary distinction is that instead of building one secondary model, four models are constructed, each tailored to its respective use case segment. This approach ensures that the specific characteristics and behaviours of each segment are effectively captured, leading to more accurate and robust FD across different contexts.

3.6. REINFORCEMENT LEARNING

RL emerges as a potential approach for optimising decision-making processes over time by leveraging recent events for adaptation. The RL experiments were motivated by the ability to prioritize the minority class, a challenge that traditional ML techniques struggle to address. Additionally, RL eliminates the need for score recalibration, often necessary in conventional ML models.

The RL framework was conceptualized to treat FD as a sequential decision-making problem. The central idea revolves around an agent that learns to classify transactions as fraudulent or genuine, with the objective of maximizing the FD while minimizing false positives. This is achieved through interactions within a custom-designed environment tailored for the classification task. This approach was inspired by the Imbalanced Classification Markov Decision Process framework formalized by Lin et al. (2019), in their study “Deep Reinforcement Learning for Imbalanced Classification”.

To implement this approach, TensorFlow Agents were used, providing a robust and flexible library for developing RL algorithms. Double Deep Q-Learning was chosen due to its improved stability and performance over traditional Q-Learning, particularly in dealing with the overestimation bias that can occur with Q-Learning (van Hasselt et al., 2015).

This approach was implemented similarly to the dynamic threshold optimisation. Each day, the RL agent was retrained with data from the previous week or two weeks and used this updated information to make decisions for that day.

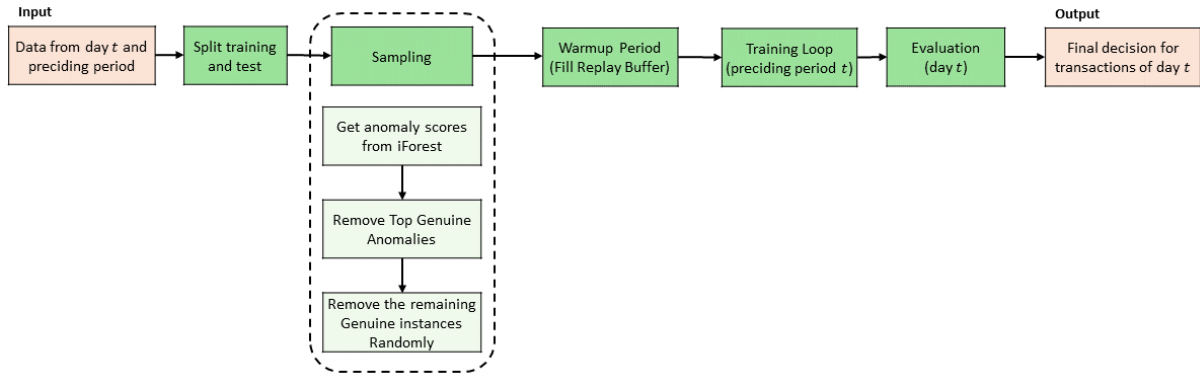


Figure 15 - Reinforcement Learning Strategy

3.6.1.1. Environment Design

The custom environment simulates the FD scenario, where the agent iterates through transaction data, making the binary classification decision. Key aspects of the environment include:

- State representation (S): Each state corresponds to a set of transaction features. The environment provides these states to the agent, representing the current transaction under consideration.
- Action space (A): The agent chooses to classify a transaction as fraudulent (1) or genuine (0). This binary action space simplifies the decision-making process.
- Transition Probability (P): The agent transitions from the current state s_t to the next state s_{t+1} following the sequence of samples in the training dataset, thus being deterministic.
- Rewards function (R): Correct classifications are rewarded positively, conversely, incorrect classifications incur penalties. In the context of imbalanced data, the absolute reward for fraudulent instances is set higher than for genuine instances. Specifically, when the agent correctly classifies a fraudulent instance, it receives a reward of +1, and a punishment of -1 for incorrect classification. For genuine instances, the agent receives a reward of λ for correct classification and a punishment of $-\lambda$ for incorrect classification, where λ is a value between 0 and 1. This approach ensures that the agent is more sensitive to the minority class, thereby improving its ability to detect fraudulent transactions.
- Discount factor (γ): $\gamma \in [0,1]$.

3.6.1.2. Training Process

A DDQN is used to estimate the Q-values, which represent the expected future rewards for taking a specific action in a given state. The training process for the DDQN agent involves several critical steps:

- **Data Preparation:** The data was undersampled using the strategy that first removes instances based on their anomaly scores and then removes additional instances randomly.
- **Network Architecture:** The DDQN is constructed with multiple layers of dense neural networks, incorporating dropout layers to mitigate overfitting. This architecture is designed to capture complex patterns in the transaction data.
- **Experience Replay:** The agent uses an experience replay buffer to store past experiences, which are sampled randomly during training. This technique helps in breaking the correlation between consecutive experiences, thereby stabilizing the training process.
- **Target Network Updates:** Periodic updates to the target Q-network ensure that the learning process remains stable. The target network provides a stable reference for updating the Q-values, preventing drastic shifts in the learned policy.
- **Warmup Period:** Before the training begins, the agent undergoes a warmup period where it collects a set number of random state-action pairs to fill the replay buffer. This initial collection helps to populate the replay buffer with diverse experiences.
- **Epsilon Decay:** To balance exploration and exploitation, an epsilon-greedy strategy is used. The exploration rate (epsilon) starts high and decays over time, encouraging the agent to explore the environment initially and gradually exploit learned policies.
- **Training Loop:** During each episode, the agent interacts with the environment, collects experiences, and periodically samples from the replay buffer to update the Q-network. Below is the pseudocode for the training loop:
 - Initialize the environment.
 - For each episode:
 - Initialize state.
 - For each time step:
 - Select action using the epsilon-greedy policy.
 - Perform action in the environment, observe reward, next state, and terminal status.
 - Store experience in the replay buffer.
 - Sample a batch of experiences from the replay buffer.
 - Calculate target Q-value.
 - Update Q-network using gradient descent.
 - Check if the episode is terminal; if yes, break.
 - Repeat.
- **Evaluation:** Simulates the Environment using the trained agent and the transactions of the day under evaluation.

3.7. EVALUATION

The primary metrics assessed in this study were the FPR and the TPR, as the main objective was to maintain the FPR at a predetermined value while maximising the TPR. In business terms, the FPR represents the false alarm rate, indicating the proportion of transactions flagged by the FDS as fraudulent that are actually genuine.

- **False Positive Rate (FPR):** Proportion of negative instances that are incorrectly classified as positive. It is calculated as the number of false positives divided by the total number of actual negative instances (Awoyemi et al., 2017).
- **True Positive Rate (TPR):** also known as recall or sensitivity, is the proportion of positive instances that are correctly classified as positive. It is calculated as the number of true positives divided by the total number of actual positive instances (Awoyemi et al., 2017).

Balancing TPR and FPR is critical in FD because it directly affects both the security and usability of financial systems. A system with a high TPR but also a high FPR would detect more fraud but at the cost of flagging many legitimate transactions, leading to customer dissatisfaction and higher costs. Conversely, a low FPR with a low TPR might mean fewer disruptions for genuine customers but at the risk of letting more fraudulent transactions go undetected. Therefore, optimizing both metrics is essential for an effective and user-friendly FDS.

In addition to FPR and TPR, other metrics were also evaluated to provide a comprehensive performance assessment. These included Precision, F1-Score, the Receiver Operating Characteristic (ROC) curve, and the Area Under the ROC Curve (AUC-ROC). Accuracy was not considered due to the high-class imbalance in the dataset.

- **ROC Curve:** The ROC curve is a graphical representation of the diagnostic ability of a binary classifier, plotting the TPR against the FPR at several threshold settings (Bradley, 1997).
- **AUC-ROC:** The AUC-ROC metric summarizes the performance of the classifier across all threshold values (the ROC curve) into a unitary values, with a higher AUC indicating better model performance (Bradley, 1997).

While metrics such as Precision, F1-Score, and Weighted F1-Score are typically valuable for evaluating classifiers on imbalanced datasets, they were not the primary focus of this study for several reasons. The F1-Score, which balances precision and recall, gives equal importance to both metrics but does not allow for explicitly fine-tuned control of FPR and TPR independently. The Weighted F1-Score, which accounts for class imbalance by giving different weights to different classes, still shares the limitation of not providing independent control over FPR and TPR. Given the need to keep FPR low while maximizing TPR, focusing on TPR and FPR aligns better with the business goals of this project.

By considering these metrics, a more holistic evaluation of the model's performance is achieved, ensuring that the detection system is not only effective in maintaining the desired FPR but also excels in other critical performance aspects.

4. RESULTS AND DISCUSSION

In this chapter, the different strategies proposed to tackle the problem and objectives discussed are assessed.

4.1. SAMPLING

This section presents the results of the three sampling strategies detailed in the methodology —random undersampling, anomaly-based undersampling, and a combined approach. The evaluation was done in a static scenario to establish a baseline understanding of how different strategies perform.

4.1.1. Isolation Forest and Extended Isolation Forest

The initial phase of our analysis focused on comparing the iForest and EIF algorithms, both of which were used to generate anomaly scores. The optimization of both models was carried out using a Bayesian hyperparameter search, by comparing the anomaly scores to the class labels, to minimize the FPR. By focusing on minimizing the FPR, this approach aimed to ensure that the false positives detected were the ones more similar to the fraudulent instances.

As explained in the methodology, the features used were conceptual variables built on business knowledge rather than typical event-based features. These variables capture the broader context of a transaction, rather than the patterns observed before and immediately after fraudulent events to give the algorithm the ability to detect anomalous instances.

To systematically evaluate the effectiveness of these models, the top 5% of instances with the highest anomaly scores were classified as fraudulent. This threshold was chosen to capture hidden frauds while minimising the removal of truly genuine instances based on anomaly scores. These predictions were then compared against the actual labels that can be observed in Table 2 - iForest and EIF Results

Table 2 - iForest and EIF Results

Model	TP	FN	FP	TN
iForest	70	212	17 029	324 687
EIF	7	275	17 092	324 624

These results demonstrated that iForest significantly outperformed EIF. The iForest algorithm not only had fewer false negatives but also achieved ten times more true positives, indicating a superior ability to correctly identify fraudulent patterns. Specifically, the iForest model exhibited a higher TPR of 0.25 compared to EIF's 0.03, and a lower FPR of 0.0498 compared to EIF's 0.0501. As a result, the iForest anomaly scores were those used in the anomaly-based and combined sampling strategies.

4.1.2. Sampling Strategies

To compare the strategies, the training set was defined for the first 8 months, and the validation set covered the last 2 months. A total of 252 experiments were conducted, testing the 3 strategies with 3 different models, 4 feature selection methods, 3 fraud rates and 5 anomaly ratios in the case of the combined strategy.

Each strategy was tested across three models (LGBM, XGB, and CatBoost) and four feature selection methods (RFE, PI, FI, and SHAP), with three different fraud ratios (0.05, 0.10, and 0.15). Additionally, the combined strategy was evaluated with five anomaly ratios (0.05, 0.10, 0.15, 0.2, and 0.3). All models underwent Bayesian hyperparameter optimization to maximize the AUC score and consider different thresholds.

The anomaly-based method, tested only with a 0.05 anomaly ratio, quickly proved ineffective as it removed genuine instances based solely on anomaly scores. This approach failed because, beyond a certain threshold, even non-anomalous instances were removed. The maximum AUC score achieved with the anomaly-based strategy was 0.68 (with the CatBoost model, PI feature selection, and a 0.1 fraud rate). In contrast, the minimum AUC score for random undersampling was 0.84, and for the combined strategy, it was 0.76. These results underscore the combined strategy's efficacy in balancing the removal of anomalies while retaining significant data.

The combined strategy emerged as the most effective approach, consistently achieving higher TPR when establishing the FPR at 0.02, and competitive AUC scores compared to random undersampling (see Table 11 - Sampling Strategies Comparison in Appendix A). The 0.02 FPR threshold was established to ensure a fair comparison among TPRs and was deemed the maximum acceptable FPR for this project.

For example, the highest AUC score (0.94) was achieved using the combined strategy with the CatBoost model, PI feature selection, and a 0.15 fraud ratio. Additionally, the highest TPR (0.51), achieved while keeping the FPR at or below 0.02, was also attained using the combined strategy with the LGBM model, PI feature selection, and a 0.15 fraud ratio.

While random undersampling did achieve better results in some configurations, the overall perspective favours the combined strategy. Overall, these results demonstrate that the combined strategy not only addresses the limitations of the anomaly-based method but also outperforms random undersampling by effectively balancing the trade-offs between removing anomalies and retaining critical information for accurate FD.

The combined strategy was designed to handle the challenges of noise and delays in fraud labelling. In many cases, fraudulent instances are not immediately marked as fraud, leading to mislabelled data. By removing instances that are similar to known fraudulent cases, the combined strategy effectively targets likely frauds that might have been marked as genuine due to these delays. After isolating these critical cases, it then randomly removes the

remaining genuine instances. This method preserves important information and enhances the model's ability to detect true fraudulent patterns, making it a more effective approach compared to random sampling.

4.2. DYNAMIC THRESHOLD OPTIMIZATION

In this section of the study, various strategies for dynamically optimizing the threshold are evaluated and discussed. This part encompasses the assessment of different approaches, including the second model approach. The evaluation focuses on determining the effectiveness of these strategies in adjusting the decision threshold every day to maximize the TPR while maintaining the FPR below the target threshold. The outcomes of these strategies are analysed to provide insights into their applicability and impact on FD efficiency.

4.2.1. Baseline Results

To evaluate the effectiveness of the implemented strategies, the metrics for the baseline model were calculated for comparison. Using only its scores, the threshold was adjusted to achieve an FPR of 0.02. This adjustment set the threshold at 0.2693, resulting in the following confusion matrix:

Table 3 - Baseline Results

		Actual Class	
		Positive	Negative
Predicted Class	Positive	620	24 969
	Negative	1 359	1 216 014

This yields an FPR of 0.02 and a TPR of 0.31. Consequently, the objective of the following strategies is to maintain or lower the FPR while achieving a higher TPR.

4.2.2. Second Model

For the second model approach, three different implementation strategies were assessed: 1) Applying the threshold directly, 2) Adjusting the threshold, and 3) Meta-Learning. Furthermore, each of these strategies could be implemented by building either one secondary model or four secondary models (one for each use case segmentation).

In the experiments for constructing the second model, to perform the sampling phase, the frequency of training the iForest model was varied to evaluate its impact on performance. Although the second model itself was trained daily, the iForest model was tested with annual training (trained once with data from the first 8 months) and weekly training (retrained weekly

using data from the previous month). This variation aimed to determine if more frequent retraining of the iForest would significantly enhance the second model's ability to detect fraudulent transactions, balancing model accuracy and timely updates.

4.2.2.1. Direct Threshold Adjustment

The first strategy assessed in this study involved directly applying the threshold derived from the secondary model to the baseline model's scores. This approach aimed to leverage the secondary model's output to dynamically adjust the threshold for FD in real time. However, it became evident after the initial trial that this strategy would not be effective.

Table 4 - Direct Threshold Adjustment Results

		Actual Class	
		Positive	Negative
Predicted Class	Positive	52	1 187
	Negative	1 576	1 121 864

This results in an FPR of 0.01, however, it also yields a TPR of 0.03 which is far lower than the reference value (0.31).

The primary issue encountered was that the outputs of the secondary model exhibited a different distribution and behaviour compared to the baseline scores. Even with the standardization of scores, this discrepancy persisted, meaning that directly applying the threshold from the secondary model to the baseline model's scores did not yield the desired results. The difference in distributions led to misalignments in the thresholding process.

Due to these challenges, it was soon realized that this strategy would not be feasible. The direct application of thresholds did not account for the inherent differences between the secondary model's outputs and the baseline model's scoring mechanism. Consequently, this approach was abandoned in favour of more sophisticated methods.

4.2.2.2. Adjusting the threshold

This strategy was evaluated in two settings, both utilizing an LGBM model with RFE feature selection, training the iForest annually, and a combined sampling strategy that removed 5% of anomalies to achieve a 5% fraud rate. The secondary model was trained with data from either the previous week or the previous two weeks. The results are summarized in the following table:

Table 5 - Performance of Weekly vs. Two-Week Training

Model	TP	FN	FP	TN
Weekly	929	1 053	56 504	1 184 206
Two weeks	1 292	687	113 530	1 127 180

Analyzing these results, it was inferred that training the secondary model with data from the previous two weeks yielded a higher TPR of 0.65, indicating better detection of fraudulent transactions. However, this came at the cost of an FPR of 0.09, which differs from the target FPR by 0.07, suggesting that this strategy might provoke an unacceptably high number of false positives, potentially leading to inefficiencies and higher operational costs.

To address this FPR issue, training the secondary model with the previous two weeks, the fraud rate was increased to 15% during the sampling phase, resulting in the following outcome:

Table 6 - Effect of Increasing Fraud Rate

TP	FN	FP	TN
714	1 265	33 343	1 207 367

This adjustment successfully reduced the FPR to 0.03, but it also resulted in a decrease in the TPR to 0.36. To further refine the approach, the training period was adjusted to one month, and the anomaly scores were used in an iForest training session conducted weekly.

Table 7 - Performance with Monthly Training and Weekly iForest Sessions

TP	FN	FP	TN
547	1 081	25 964	1 097 087

This adjustment achieved an FPR of 0.023, which is close to the target, but resulted in a TPR of 0.34, lower than previous attempts.

This strategy did not yield significantly better results compared to those without optimization. However, it highlighted the potential for developing more robust approaches to consistently achieve the target FPR, allowing a primary focus on maximizing the TPR.

4.2.2.3. Meta-Learning

In the context of adjusting the threshold using a secondary ML model, meta-learning emerges as an effective strategy to achieve the desired objective. By adjusting the threshold of the

meta-learning model directly, it is guaranteed to achieve the desired FPR (0.02). This level of control over the FPR allows the focus to shift towards adjusting other settings to achieve the highest possible TPR.

Several experiments were conducted to evaluate this strategy. The meta-learning model was initially tested using only two input variables: the scores from the baseline model and the secondary model. Subsequently, additional variables were incorporated into the model. These additional variables differed from those built in the feature engineering phase, as they conveyed information about potential fraud but without directly representing the label. For example, the additional variables included information about actions such as blocking, or cancellation associated with the card involved in the transaction during a period preceding the transaction.

All experiments achieved an FPR of less than 0.02, demonstrating not only the ability to control the FPR but also to lower it in relation to the maximum that is desired. The vast majority also demonstrated a higher TPR compared to the baseline results. It was observed that the only experiments failing to exceed the baseline TPR (0.31) were those where the secondary model was trained with data from the week preceding the test days. This outcome is attributed to the limited number of fraudulent instances in the dataset. A longer training period provides the model with a more diverse set of instances to learn from, leading to better results. However, the training period was capped at one month to ensure that the model could still learn from recent data, striking a balance between having enough data for learning and maintaining the relevance of the data.

In this sequence, the best experiences achieved were those in which the secondary model was trained with data from one month preceding the test days. The following table shows the best results:

Table 8 - Meta-Learning Results

iForest regularity	Feature Selection	Fraud Rate	Anomaly Ratio	# Variables	TP	FN	FP	TN	TPR	FPR
annually	RFE	0.05	0.05	2	911	1 068	20 912	1 219 798	0.46	0.017
annually	PI	0.15	0.10	2	908	1 071	21 278	1 219 432	0.46	0.017
weekly	RFE	0.05	0.05	4	716	912	18 834	1 104 217	0.46	0.017
weekly	RFE	0.05	0.05	2	696	932	18 025	1 105 026	0.43	0.016

The first three experiments in the table achieved the highest TPRs of 0.46 with an FPR of 0.017. The last experiment displayed a slightly lower TPR of 0.43, accompanied by a marginally lower FPR of 0.016. Nonetheless, all experiments successfully met the objective of surpassing the baseline TPR while maintaining an FPR below the established threshold.

The meta-learning strategy proves to be an effective approach for dynamic threshold adjustment in FD. By leveraging the strengths of a secondary model and incorporating a variety of informative features, it is possible to improve the balance between minimizing false positives and maximizing the detection of fraudulent activities.

4.2.2.4. Segment-Specific Strategy

In a final attempt to improve the results with dynamic threshold optimization, a secondary model was applied to each of the segments identified in the data exploration phase. Using the meta-learning approach with a LGBM as the secondary model, feature selection RFE, Fraud Rate of 0.05, Anomaly Ratio of 0.05, and training iForest annually, it was possible to obtain the following results:

Table 9 - Segment-Specific Strategy Results

TP	FN	FP	TN
981	988	7 452	1 233 258

These numbers translate into a TPR of 0.50 with an FPR of 0.006. These were the best results achieved using the second model to dynamically optimize the threshold, which not only presented the highest TPR but also managed to reduce the FPR to less than half of the intended limit.

These findings emphasize the importance of understanding the data and business context when developing effective FDS. Creative approaches that diverge from standard methods can greatly enhance outcomes, showcasing the value of innovative thinking in achieving superior results.

4.2.2.5. Comparative Analysis

Based on the results presented in this section, several conclusions can be drawn beyond identifying the best outcomes.

Training the iForest to obtain anomaly scores for use in the sampling phase deviated somewhat from the logic of training the secondary model daily. The different training frequencies for the iForest did not prove significantly relevant, indicating that it is unnecessary to perform this task daily in conjunction with the secondary model. Therefore, within the scope of the strategies demonstrated here, it is not pertinent to expend computational resources on daily training of the iForest.

Regarding the fraud ratio in the samples used to train the secondary model, higher ratios such as 0.15 showed better results in controlling the number of false positives. However, this also led to a lower number of true positives. Thus, using a fraud ratio of 0.05 was preferred to increase the number of true positives when effectively fixing the false positive rate (FPR).

In terms of the various model settings tested, the different feature selection techniques did not prove highly relevant. Nevertheless, Recursive Feature Elimination (RFE) was used in the experiments with the best results. However, RFE was also the most frequently used method, making it difficult to draw a definitive conclusion.

As previously mentioned, using the secondary model to adjust the threshold of the baseline model is not the optimal strategy. Despite all efforts, there is no control over the scores of the main model, and it is challenging to consistently fix the FPR to the desired level.

Consequently, the strategy of using a third model (meta-learning), which allows for direct control over the FPR, presents greater potential. By directly controlling its threshold, it is possible to achieve the desired outcomes.

Furthermore, segmenting the construction of the second model allowed for learning different fraud patterns. Although these patterns occur within the same timeframe, they are distinct. Therefore, building models for each segment enabled better capture and prediction of these patterns.

4.3. REINFORCEMENT LEARNING

In this section of the study, the assessment focuses on the results obtained using the RL-based approach. The experiments in this section were carried out over a shorter period of time due to the need for computer resources. In any case, TRP and FPR can be used as metrics for comparison with the threshold dynamic optimization method. The best results were the following:

Table 10 - Reinforcement Learning Results

Experiment	TP	FN	FP	TN
1	76	115	46820	126506
2	24	178	3254	170072

Both experiments were set to run for 100 000 episodes, with 1 000 warmup steps to collect data using a random policy. The memory length was also set to 1 000, matching the number of warmup steps. The batch size for training the model was 32, and 2 000 steps were collected per episode, with data collection occurring every 500 steps. The learning rate, which determines the step size during the gradient descent update, was 0.0002. The discount factor, gamma, balances immediate and future rewards and was set to 0.7.

To balance exploration and exploitation, the minimum epsilon value was set to 0.5, representing the minimal chance of choosing a random action, and the epsilon decay was configured to reduce from 1.0 to this minimum value over 10 000 episodes.

The difference between the experiments lies in the sample used for training. Both were undersampled with the combined strategy, removing 5% of the anomalies, but in experiment 1 the fraud rate was set at 1%, and in the second the fraud rate was set at 5%.

Experiment one corresponds to a TPR of 0.42 and a FPR of 0.283, and experiment 2 corresponds TPR of 0.14 and a FPR of 0.02. Both fall a little short of the results obtained through dynamic threshold optimization.

RL was a complex strategy to implement, and its application in the context of this FD data remains somewhat enigmatic. Nevertheless, valuable insights into its workings were gained.

Balancing the sampling has a significant influence. Removing more genuine transactions from the sample results in an increased number of false positives, highlighting the importance of carefully managing the balance between genuine and fraudulent instances to maintain optimal performance.

Removing scores and thresholds, and making a discrete decision offers several advantages in a FDS. One primary advantage is the simplification of the decision-making process, which enhances the system's interpretability and operational efficiency. By making a direct classification decision (fraudulent or genuine) rather than relying on a risk score and a threshold, the system avoids the complexity of calibrating thresholds that can vary over time and require continuous adjustment. However, this approach also has its disadvantages. In this specific context, establishing a precise threshold to achieve the desired FPR and maximize the TPR is not feasible. This limitation means the process is somewhat reliant on the overall optimization of results, rather than being able to specifically focus on maximizing the TPR while maintaining a stable FPR.

There remains a considerable difficulty with applying deep learning to tabular data. In the current context of FD, this method is very complex and resource-intensive, whereas simpler ML models can effectively complete the task. Nevertheless, as fraud patterns continuously evolve, RL holds significant potential for future exploration, providing a dynamic and adaptable approach to FD.

5. CONCLUSIONS AND FUTURE WORK

FD remains a critical challenge in the dynamic financial ecosystem. With the proliferation of fintech solutions, fraud mechanisms have become increasingly sophisticated, necessitating innovative approaches to effectively identify and mitigate fraudulent activities. Traditional FD methods often suffer from high costs, imprecision, and time consumption, underscoring the need for advanced ML techniques. This project addresses these challenges by exploring dynamic threshold optimization and RL to improve the performance and adaptability of FDS.

The primary challenge addressed in this study is the dynamic nature of FD, which involves balancing the trade-off between false positives and false negatives. This research proposes dynamic threshold optimization and RL-based techniques to enhance the robustness and accuracy of FDS. The project used real-world data from a major European transaction payment processor, ensuring the solutions are grounded in practical applicability.

The project followed a CASP-DM methodology, which proposes making the model/system versatile. This adaptability is crucial for maintaining the relevance and effectiveness of the model in the face of changing fraud tactics. The methodology is structured around key phases: understanding the business context, exploring available data, creating new features, defining the sampling strategy, and developing the proposed strategies. The proposed strategies were essentially divided into two categories, dynamic threshold optimization, where a secondary model trained with more recent data was trained every day to define daily thresholds, and a RL-based approach that challenges the current FDSs, eliminating the need for risk scores and thresholds.

To address the overarching objective of enhancing FDS through dynamic optimization techniques, this research was guided by several specific research questions:

How can the impact of data drift and concept drift on the performance of fraud detection models be mitigated?

To address this, the study employed undersampling with a focus on a robust approach that selects specific instances to remove and then applies random sampling. Additionally, dynamic and versatile techniques were implemented to ensure the model remains effective over time.

What sophisticated sampling methods can address data imbalance in fraud detection, and what are their impacts on model performance?

A new proposed undersampling strategy was performed, which involved removing selected instances based on anomaly scores given by an unsupervised iForest algorithm and then removing the remaining instances randomly. This method proved effective in addressing data imbalance and improving model performance.

How can the threshold for FDS be dynamically optimized to balance the trade-off between false positives and false negatives effectively?

A second ML model was trained using recent data to capture emerging fraud patterns, and a meta-learning model was constructed with dynamically fine-tuned thresholds. The meta-model was designed to take output scores from both the baseline and secondary models as inputs, allowing it to adjust thresholds based on evolving data patterns. This approach ensures that validation data is processed through both the secondary model and the meta-model, enhancing the system's adaptability and accuracy in FD.

How can a FDS be changed to eliminate the decision-making process based on risk scores and thresholds?

A RL technique was employed. Although RL is more commonly used in use cases of computer vision, it has shown great potential in the context of tabular data where dynamic patterns frequently occur. This approach allows the FDS to make direct accept/reject decisions based on contextual data and the RL agent's continuous learning capabilities, thus eliminating the reliance on predefined thresholds.

5.1. WORK CONTRIBUTION

This research makes significant contributions to the field of FD, both practically and theoretically, by addressing critical gaps in current methodologies and providing innovative solutions to persistent challenges.

Firstly, it presents a robust approach to dynamically optimizing the performance of real-world ML models. This includes the development and refinement of various strategies aimed at enhancing the adaptability and effectiveness of FDS. The study particularly focuses on addressing the challenges posed by imbalanced data through advanced sampling techniques, thereby improving the performance and accuracy of the ML models used in the proposed strategies.

Moreover, this research demonstrates how to enhance the results of an ML model without altering the underlying algorithm. By implementing a parallel dynamic technique based on the use of a second ML model, and implementing RL approaches, it offers practical solutions for real-time FD, ensuring that the models remain effective even as fraud patterns evolve.

The contributions of this research are not only practical but also extend the current body of literature, which is often limited in terms of real-world data applications. This study provides empirical evidence and insights that are crucial for the ongoing development of effective FDS, helping the field to keep pace with the rapidly changing landscape of financial transactions and fraud mechanisms.

5.2. CONSTRAINTS AND LIMITATIONS

This project was developed using data from a major transaction payment processor in Europe. The access and use of this data had to comply with the company's security policy.

Furthermore, to develop the proposed solutions, the company's IT resources were used. The resources were shared with the working team and their usage had to be managed carefully to avoid disrupting the daily operations of the company's employees who rely on the same infrastructure.

5.3. FUTURE WORK

In light of the findings and methodologies presented in this thesis, several avenues for future work can be identified to further enhance the dynamic optimization of FDS.

Investing more in the logic of variable creation and feature engineering is a key area for improvement. Exploring different variables and incorporating more contextual information can significantly bolster the model's robustness.

Combining multiple models in a hybrid approach could also prove beneficial. In this project, unsupervised learning was primarily used in the sampling phase, but there is potential to extract more value from it. For instance, anomaly scores could be used as input variables in a supervised model, enhancing its predictive power.

In terms of sampling, an alternative to the unsupervised approach used for removing transactions labelled as genuine is to use a self-supervised learning methodology, which relies only on transactions labelled as fraud and those confirmed as genuine by clients (H. Liu et al., 2022). Through multiple iterations of this method, transactions with high model scores but not labelled as fraud are removed. In subsequent iterations, only confirmed genuine transactions and those not confirmed by clients but with a low-risk score are incorporated.

Additionally, depending on the dimensionality of the feature space, a nearest neighbour algorithm can be employed to propagate the confirmed genuine transactions. However, this approach carries several risks: besides the computational costs, it may incorrectly label some patterns associated with social engineering fraud or malware as genuine transactions. Moreover, during daily training, there is a lack of immediate feedback from clients.

The application of RL for decision-making is another promising area for further development. The initial work demonstrated here with DDQNs can be expanded and refined. Future research could explore more advanced RL algorithms, such as Proximal Policy Optimization or Deep Deterministic Policy Gradient, to enhance the agent's decision-making capabilities. Additionally, experimenting with different environments and reward structures within the RL framework can help better capture the complexities and nuances of real-world FD scenarios. Integrating contextual information and business-specific insights into the RL model can further increase its robustness and adaptability. Moreover, as was done for the second model, RL can

also be explored for optimizing thresholds dynamically, potentially offering a sophisticated way to balance the trade-off between false positives and false negatives more effectively.

Exploring RL with other models, such as tree-based methods instead of neural networks, could provide more interpretable and explainable results. This approach, as suggested in recent studies, would enhance the transparency of decision-making processes (Min & Elliott, 2022).

Despite the significant efforts made in this project, there is ample room for improvement. This thesis opens the door to numerous new ideas and experiments. The continuous evolution of fraud tactics necessitates a dynamic and adaptable approach to FD. By leveraging a combination of advanced ML techniques and RL, future research can aim to build more resilient, accurate, and responsive FDS.

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APPENDIX A

Table 11 - Sampling Strategies Comparison

	Model	Feature Selection Method	Fraud Rate	Sampling Strategy with highest TPR	TPR	Sampling Strategy with highest AUC	AUC
1	catboost	FI	0.05	combined_0.1	0.29	combined_0.15	0.92
2	catboost	FI	0.1	combined_0.1	0.37	combined_0.05	0.92
3	catboost	FI	0.15	combined_0.15	0.47	combined_0.15	0.92
4	catboost	PI	0.05	combined_0.3	0.31	combined_0.05	0.92
5	catboost	PI	0.1	combined_0.3	0.41	random_undersampling	0.92
6	catboost	PI	0.15	combined_0.15	0.47	combined_0.15	0.94
7	catboost	RFE	0.05	random_undersampling	0.24	combined_0.1	0.88
8	catboost	RFE	0.1	random_undersampling	0.29	combined_0.05	0.86
9	catboost	RFE	0.15	combined_0.1	0.37	combined_0.05	0.87
10	catboost	SHAP	0.05	combined_0.1	0.31	combined_0.1	0.92
11	catboost	SHAP	0.1	combined_0.05	0.39	combined_0.15	0.92
12	catboost	SHAP	0.15	random_undersampling	0.43	random_undersampling	0.93
13	lgbm	FI	0.05	combined_0.1	0.28	random_undersampling	0.93
14	lgbm	FI	0.1	random_undersampling	0.39	combined_0.15	0.93
15	lgbm	FI	0,15	random_undersampling	0.49	random_undersampling	0.93
16	lgbm	PI	0.05	combined_0.3	0.27	combined_0.1	0.93
17	lgbm	PI	0.1	combined_0.2	0.41	random_undersampling	0.92
18	lgbm	PI	0.15	combined_0.1	0.51	combined_0.1	0.92
19	lgbm	RFE	0.05	random_undersampling	0.23	combined_0.05	0.88
20	lgbm	RFE	0.1	combined_0.1	0.28	combined_0.05	0.88
21	lgbm	RFE	0.15	combined_0.1	0.34	random_undersampling	0.89
22	lgbm	SHAP	0.05	random_undersampling	0.24	combined_0.05	0.92
23	lgbm	SHAP	0.1	combined_0.05	0.37	random_undersampling	0.92
24	lgbm	SHAP	0.15	-	-	combined_0.05	0.92
25	xgb	FI	0.05	combined_0.05	0.29	combined_0.05	0.92
26	xgb	FI	0.1	combined_0.05	0.41	combined_0.05	0.92
27	xgb	FI	0.15	combined_0.1	0.49	combined_0.1	0.92
28	xgb	PI	0.05	combined_0.3	0.34	random_undersampling	0.92
29	xgb	PI	0.1	combined_0.1	0.42	random_undersampling	0.92
30	xgb	PI	0.15	combined_0.2	0.49	combined_0.2	0.93
31	xgb	RFE	0.05	random_undersampling	0.22	combined_0.1	0.87
32	xgb	RFE	0.1	random_undersampling	0.31	combined_0.15	0.88
33	xgb	RFE	0.15	combined_0.1	0.36	random_undersampling	0.86
34	xgb	SHAP	0.05	random_undersampling	0.27	combined_0.1	0.92
35	xgb	SHAP	0.1	random_undersampling	0.39	combined_0.05	0.92
36	xgb	SHAP	0.15	-	-	random_undersampling	0.92

